

Forward Modelling the ASKAP-FLASH 21cm Absorption Survey with SANGRIA: Geometry Alone Cannot Explain High HI Detection Fractions Towards Compact Radio Sources

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ABSTRACT

The 21 cm hydrogen line is an important probe of the cool, neutral medium in and around galaxies, which is crucial to galaxy formation and evolution. However, a number of factors influence the detectability of HI in absorption beyond just the total gas mass of the system. We investigate here two such parameters: galaxy geometry and radio morphology, using a suite of mock observations produced using the SANGRIA (Simulating Absorption of Neutral Gas for Radio Astronomy) code, with galaxy data drawn from the SIMBA cosmological simulations. These mocks are matched to the observational parameters of the First Large Absorption Survey in HI (ASKAP-FLASH), an all-sky, untargeted search for HI absorption currently being conducted with the Australian SKA Pathfinder. We focus on instances of associated absorption, where the continuum backlight is produced by the jets of an active galactic nucleus associated with the same galaxy hosting the HI, so that we can investigate the apparent over-detection of HI towards galaxies with compact radio jets ($\lesssim 1$ kpc) in ASKAP-FLASH. We find that, although simple geometric arguments can explain some of the preferential detection towards compact jets, they are not sufficient to explain the high detection rates seen in the ASKAP-FLASH Pilot Surveys. We interpret this discrepancy between the geometrically-driven detection rate and the observed detection rate towards compact sources to be evidence that the most compact AGN jets reside in uniquely gas-rich environments.

Keywords: Galaxies: active (17) — Radio Jets (1347) — Neutral atomic hydrogen (1099) — Methods: computational (293)

1. INTRODUCTION

Neutral atomic hydrogen (HI) is a fundamental ingredient in star formation and galaxy evolution, and is crucial to understanding the cosmic baryon cycle (C. Péroux & J. C. Howk 2020). Fortunately, it is detectable in absorption out to arbitrarily high redshifts (provided the radio continuum backlight is sufficiently bright), making it an excellent tracer of how the interstellar medium (ISM) evolves over time (R. Morganti & T. Oosterloo 2018). For this reason, the First Large Absorption Survey in HI was born (ASKAP-FLASH; J. R. Allison et al. 2022; H. Yoon et al. 2025), to trace the evolution of HI in the under-studied period after cosmic noon ($z = 0.4-1$) where a marked decline in star formation rate density is observed, likely caused by depleted cold gas reservoirs coupled with feedback mechanisms

(P. Madau & M. Dickinson 2014). ASKAP-FLASH is one of nine Survey Science Projects being carried out by the Australian SKA Pathfinder radio telescope (A. W. Hotan et al. 2021), and is expected to complete its observations of the southern radio sky by mid-2026.

While the ASKAP-FLASH survey is ongoing, a trend is emerging amongst the HI-detected sources so far, in which there are an over-representation of compact “Peaked Spectrum” (PS) radio AGN compared to their number amongst the general radio population (E. F. Kerrison et al. 2024; H. Yoon et al. 2025). This PS classification is based on the fact that the radio emission in these objects peaks at some characteristic frequency (proportional to the size of the jet), before turning over at low frequencies due to absorption losses. PS sources are thus believed to be young or confined AGN, with their radio jets extending out to only a few kiloparsecs, making them intimately connected with the ISM of the host galaxy (C. P. O’Dea & D. J. Saikia 2021). Al-

ready in targeted searches for H I absorption it has been noted that PS sources exhibit a high detection rate of H I (e.g. J. Conway 1997; K. Geréb et al. 2014), but ASKAP-FLASH is the first time such a skew has been seen in a completely untargeted search of extragalactic radio sources. This presents an interesting puzzle; are these recently-triggered radio AGN inherently more likely to host the cool, neutral medium detectable in H I absorption, or is this trend simply the product of a geometrical selection effect related to their extreme compactness? One avenue for testing these hypotheses is to use mock observations of simulated galaxies, where the underlying distribution of the gas is known exactly.

Mock observations (or mocks), where simulated datasets are post-processed to produce observable quantities, are a well-established tool for exploring the complexities of galaxy evolution and dynamics. At radio wavelengths, mocks of H I 21 cm emission maps have been used to study resolved rotation curves (F. A. Roper et al. 2023) and line profile asymmetries (M. Glowacki et al. 2022), drawing data from the SIMBA cosmological simulations (R. Davé et al. 2019). Other studies have used post-processing to examine the global properties of 21 cm emission in IllustrisTNG (A. R. H. Stevens et al. 2019) and EAGLE (R. A. Crain et al. 2017), and K. El-Badry et al. (2018) demonstrated that gas kinematics could be derived even from unresolved H I profiles using the FIRE simulations (P. F. Hopkins et al. 2018). At even higher resolution, C. M. Booth & T. Theuns (2007) studied 21 cm emission mocks from idealised simulations to understand the role of H I in galactic fountains. F. Sinigaglia et al. (2022) similarly used mock observations to better understand the systematics involved in spectral stacking, a technique used to recover statistical measures of H I-mass in large samples of galaxies. However, these H I studies remain focused on emission in the local Universe. At much higher redshifts, P. M. Geil & J. S. B. Wyithe (2008) used a semi-numerical approach to understand how sensitive the current generation of low frequency instruments might be to H I emission at the epoch of reionisation, and others have been working on mock catalogues and tools to predict the detectability of H I with the SKA (e.g. D. Obreschkow & M. Meyer 2014; E. C. Elson et al. 2016). Still, all the work in this space has focused heavily on H I emission, primarily at low $z \lesssim 0.2$ and high $z \gtrsim 6$ redshift. There remains a large fraction of cosmic history for which we have no H I mocks because we have no observational data to compare them to; it is observationally difficult (indeed, impossible with current instruments) to detect H I in emission at these redshifts. However, absorption studies are currently able to observe H I out to $z = 3.5$

(J. N. H. S. Aditya et al. 2021), and a number of surveys have been designed to search for H I in this way towards lists of pre-defined targets spanning various slices of redshift space including $0.7 < z < 1$ (with 29 targets; S. Murthy et al. 2022), $0 < z < 2$ (with 391 primary targets; N. Gupta et al. 2016), $0 < z < 2.74$ (with 260 targets; K. Grasha et al. 2020), and $0.8 < z < 3.5$ (with 22 targets; N. Kanekar et al. 2014). ASKAP-FLASH is the only existing survey to search for H I absorption in an untargeted manner towards *all* radio continuum sources (above a flux density limit $S \geq 30$ mJy) across a large ($24,000 \text{ deg}^2$) region of sky and a wide redshift range ($0.4 < z < 1.0$). It is for this reason that we now have suitable datasets to compare with mock observations of H I absorption at cosmic noon.

In this paper, we present a first analysis of neutral hydrogen probed in absorption by the 21 cm line in a cosmological simulation. Using the Simulated Absorption of Neutral Gas for RadIo Astronomy package (SANGRIA; Ward, Kerrison & Tonnesen *in prep.*), we inject radio jets into galaxies from SIMBA which serve as a backlight for the hydrogen permeating the galaxy (Section 2). We develop three physically-motivated jet models (Sections 2.2 & 3.3) which are injected into galaxies in order to create mock spectral line observations (Section 2.4). Our suite of mock observations is drawn from a representative sample of gas-rich galaxies originating in a single SIMBA snapshot (Section 3). With this suite we test the effect of varying the radio jet parameters (Sections 4.1 & 4.2), the orientation of the host galaxies (Section 4.3), and the orientation of the radio jets (Section 4.4) using a uniform parameter search. We also consider the effect of galaxy morphology on our ability to detect H I (Section 4.5). Finally, we re-weight our uniform parameter search to better match the observed luminosity function of radio galaxies, in order to determine how much of the high detection rate towards compact radio AGN can be attributed to geometry alone (Section 5). We discuss our results and present our conclusions in Sections 6 & 7.

Throughout this work we assume a Λ CDM cosmology with parameters of $H_0 = 68 \text{ km s}^{-1} \text{ Mpc}^{-1}$, $\Omega_M = 0.3$, $\Omega_\Lambda = 0.7$ and $\Omega_b = 0.048$, following Planck Collaboration XIII (2016).

2. RADIO OBSERVATIONS OF COSMOLOGICAL SIMULATIONS

In this section, we outline the basic methodology behind the SANGRIA pipeline for producing mock H I absorption spectra. Further details will be described in a forthcoming companion paper (Ward, Kerrison & Tonnesen *in prep.*). Figure 1 shows an overview of the

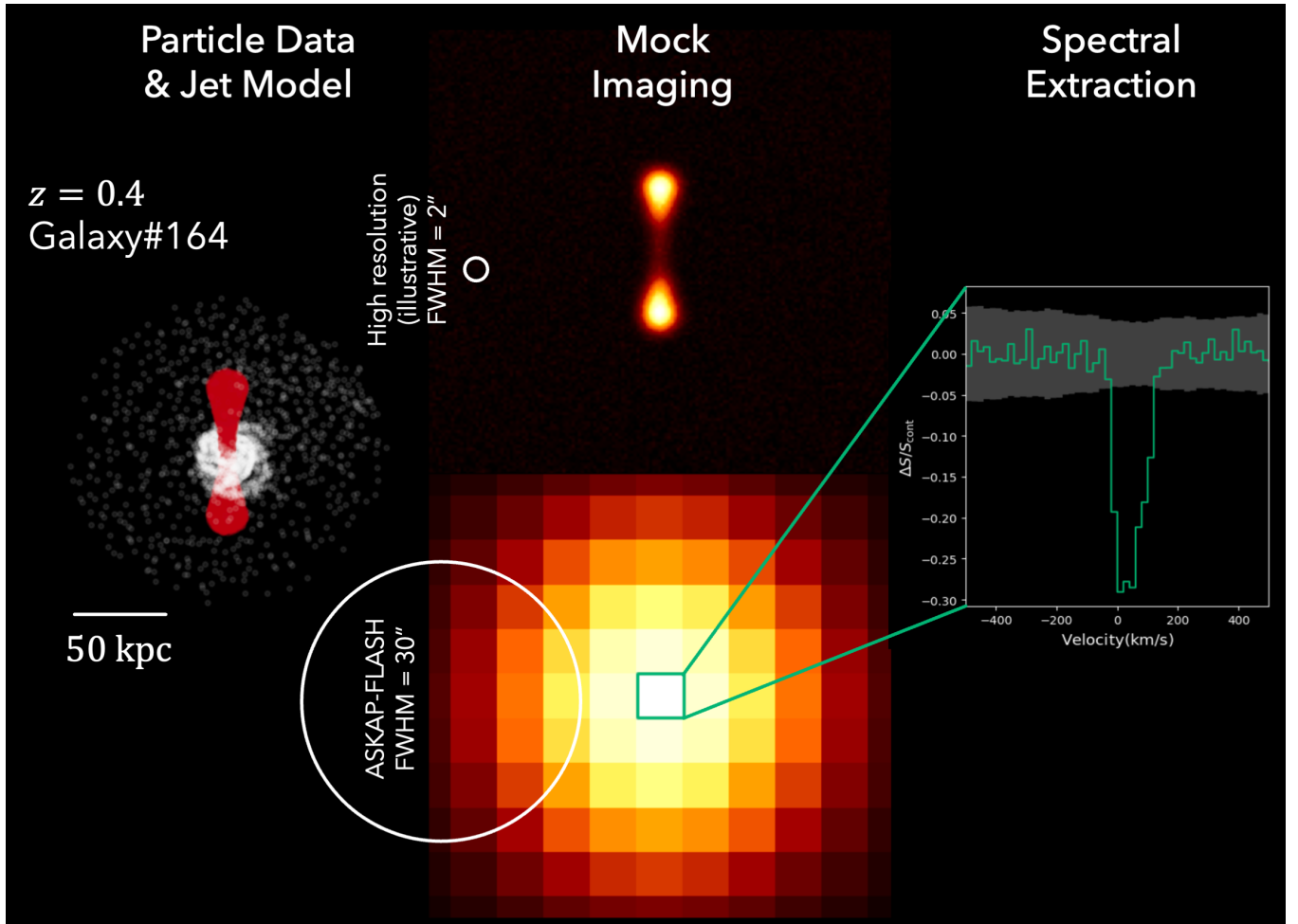


Figure 1. An outline of the SANGRIa pipeline, translating particle and ray data into mock H I spectra. *Left:* the SIMBA particle data (white points) from Galaxy #164 and our injected jet model (red points). *Middle:* mock radio observations of this system after beam convolution. We show two beam sizes: *top* an illustrative high resolution FWHM = 2'' beam and *bottom* the ASKAP-FLASH beam (30''). *Right:* the H I spectrum extracted from the brightest pixel of the ASKAP-FLASH spectral cube, showing a clear 21cm absorption line.

pipeline – on the left we show the gas particle data from a SIMBA galaxy (see Section 2.1) and an injected FRII jet model (Section 2.2). The middle panels show mock observations of this system (Sections 2.3 & 2.4), showing both a FWHM = 30'' beam matched to the parameters of the ASKAP-FLASH survey (bottom), and a high resolution beam of FWHM = 2'' (top) for illustrative purposes. The right column shows the radio spectrum around the 21cm H I line extracted from the brightest pixel. The following sections will now describe these steps in more detail.

2.1. The SIMBA simulations

The parameters of the ASKAP-FLASH survey place constraints on many of the characteristics required of our chosen simulation; it must have global gas masses well-matched to observations not only in the local Universe, but at intermediate redshifts ($0.4 < z < 1.0$), it

should implement some well-motivated AGN formation, growth and feedback prescription (in order to produce galaxies likely to host radio jets), and it should contain a diverse population of galaxies including both mergers and isolated systems.

We construct our mock observations from the SIMBA⁴ cosmological simulations (R. Davé et al. 2019), run with the meshless finite mass hydrodynamics solver from GIZMO (P. F. Hopkins 2015). We use the fiducial $m100n1024$ run, which has a comoving box size of $L = 147$ cMpc and a baryonic mass resolution of $m_b = 1.83 \times 10^7 M_\odot$.

Gas cooling and photoionization heating are computed using the GRACKLE-3.1 library (B. D. Smith et al. 2017), assuming a spatially uniform UV background and in-

⁴ <http://simba.roe.ac.uk>

cluding self-shielding following [A. Rahmati et al. \(2013\)](#). As cosmological simulations do not resolve the cold neutral medium, the neutral and molecular hydrogen fractions are estimated using subgrid prescriptions ([M. R. Krumholz & N. Y. Gnedin 2011](#)). The resulting H I and H₂ masses are calculated on-the-fly for each gas particle and stored in the snapshots, which we use directly in our ray-tracing procedure, as explained in the next section.

These simulations include a bi-directional, hydrodynamically-decoupled outflow model at low Eddington ratios ($f_{\text{Edd}} < 0.2$) to approximate ‘jet-mode’ AGN feedback, and per-particle H I masses can be read directly from SIMBA snapshot files. Previous work has also shown that with sensible selection criteria and prescriptions for radio luminosity, SIMBA samples can be well-matched to observational samples of radio galaxies in the local Universe ([N. Thomas et al. 2021](#); [N. Thomas & R. Davé 2022](#); [N. Thomas et al. 2025](#)), making an apples-to-apples comparison with ASKAP-FLASH data readily achievable using these simulations.

Previous work has introduced mock observation pipelines for studying neutral hydrogen in emission (e.g. [MARTINI, K. A. Oman 2019](#); [K. A. Oman et al. 2019](#); [K. A. Oman 2024](#)). However, to study H I in absorption, additional models are needed for the background emission and radiative transfer. Since SIMBA does not directly model the luminosity or extent of the radio jets produced by the AGN, we must inject jets of appropriate size and luminosity before we can perform any analysis of 21 cm absorption. Motivated by this need for simple, flexible radio jet models within a larger mock observing framework, we have created SANGRIA (Simulating Absorption of Neutral Gas for Radio Astronomy), a robust end-to-end pipeline for producing mock 21 cm absorption line observations. While our software package has been built for flexibility and will be described in detail in [Ward, Kerrison & Tonnesen \(in prep\)](#), here we focus on one use-case: H I absorption against radio jets embedded in the hydrogen gas, matched to the parameters of the ASKAP-FLASH survey. In the following subsections we outline the methods for incorporating radio jets into the simulation, and then for extracting mock spectral line profiles from the associated galaxies.

We chose to use SIMBA in this study for the reasons outlined above. In principle though, any hydrodynamic galaxy simulation could be used by the SANGRIA pipeline that has saved per-particle H I masses, or which can be post-processed to obtain them. Expanding this study to other simulations is left for future work.

2.2. Radio Jets with SANGRIA

For an individual galaxy extracted from a SIMBA box, SANGRIA can flexibly insert one of three different jet morphologies at any arbitrary orientation and position relative to that galaxy, or the user may choose a point-like ‘quasar’ source of radio emission. For this work, we fix the origin of our jets at the centre of the galaxy, defined as the local minimum in the galactic potential, and for simplicity we only rotate the jet structure along a plane through each galaxy rather than in any arbitrary direction. The jets themselves can take one of three functional forms, depicted in Figure 2.

These models were chosen to map onto the main morphological classes of observed radio jets. The centre panel represents so-called ‘core-dominated’ radio sources with prominent jets that fade as a function of distance from the core, while the right panel represents ‘lobe-dominated’ emissions seen in large radio galaxies like Cygnus A. These are typically referred to as FR-I and FR-II galaxies respectively, after the authors of the original paper highlighting the dichotomy ([B. L. Fanaroff & J. M. Riley 1974](#)). Radio ‘emission’ is distributed along these extended jet structures according to a radial power law $L(r) \propto r^a$, where $a = 0.8$ for FRI-type jets, and $a = 2.0$ for FRII. These values for a were manually selected to produced realistic-looking jets over a range of lengths and bolometric luminosities. We were unable to tie our choices to stronger observational constraints in this case, since the average radial emission profile of jets has only been measured observationally on sub-parsec scales, which are unresolved in our setup (e.g. [E. V. Kravchenko et al. 2025](#)). However, the seminal theoretical paper on AGN jet structures also assumes a power law scaling of this kind ([R. D. Blandford & A. Königl 1979](#)).

Our third model represents the more recent FR-0 classification describing extremely compact radio AGN ([R. D. Baldi et al. 2015](#)), for which we use the model in the left-hand panel of Figure 2. Very Long Baseline Interferometry shows that many FR-0 sources are unresolved even on milliarcsecond scales ([R. D. Baldi et al. 2025](#)). Given that even in the local $z \leq 0.1$ Universe this corresponds to an angular scale of less than 1 arc-second, these will be entirely unresolved in our mock images with matched ASKAP-FLASH resolutions. A simple, spherical emission model like that chosen here is therefore an appropriate representation of these most compact jets.

Both the FR-I and FR-II models are formed from bi-conical jets centred at the origin, while the FR-0 is a sphere. Within the three dimensional jet structure, the luminosity is discretised into individual rays distributed

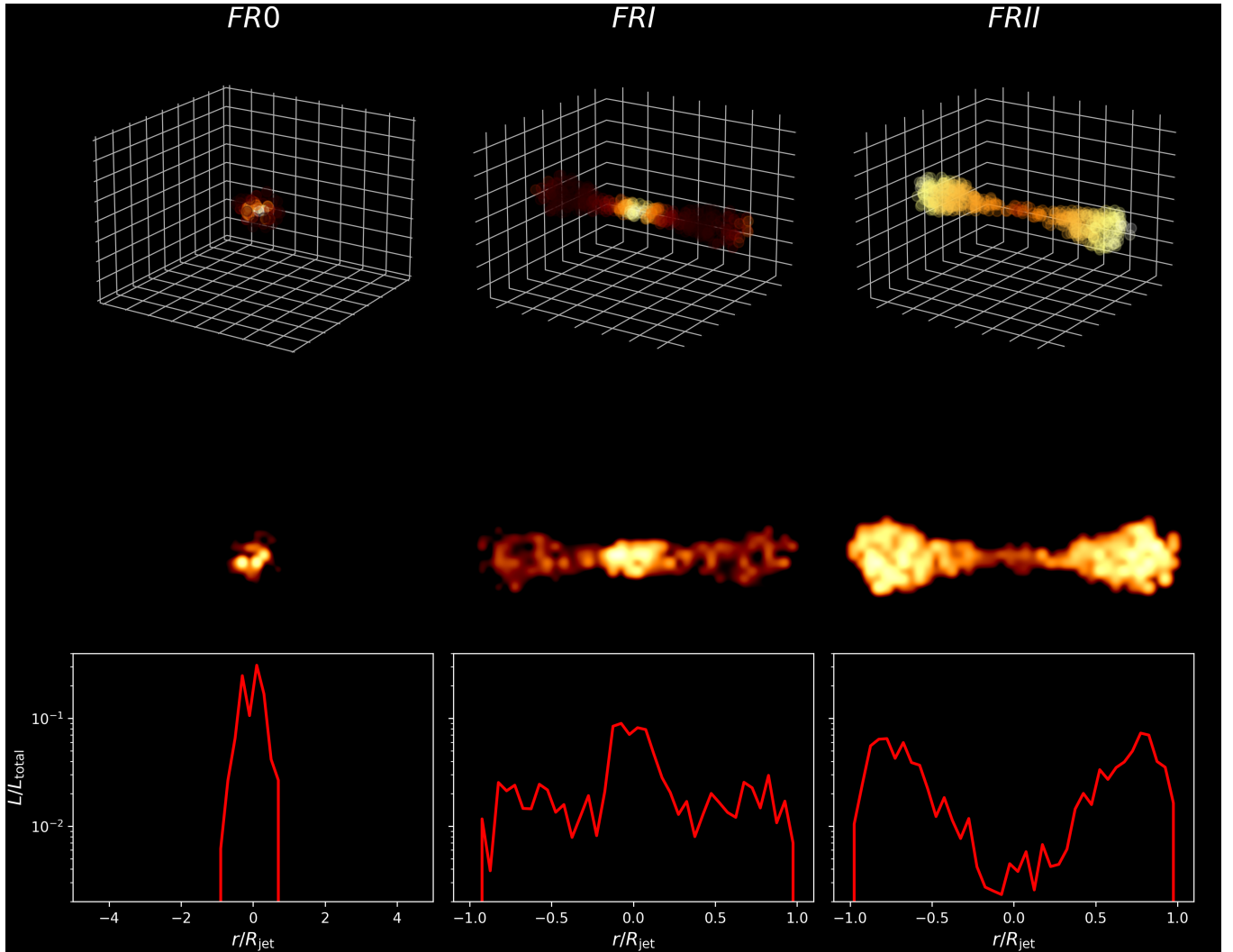


Figure 2. The luminosity distribution for the three jet types used in this work. These map onto the observational classes of FR0 or compact (left), FR-I or core-dominated (centre) and FR-II or lobe-dominated (right). The sizes and luminosities of these jet models are scaled to the required values before they are added into the SIMBA particle data and used as the backlight for the absorption ray tracing.

efficiently across the jets using a glass packing algorithm, which also allows us to emulate some of the irregularities of jet structures seen observationally as hot spots and knots (e.g. [G. Miley 1980](#)). Our results are insensitive to the number of jet points used, as long as the local number density of jet particles is greater than the local number density of gas particles to ensure we do not undersample the gas, especially in the higher density disk.

2.3. Ray tracing

Each jet particle is used as the source point for ray casting. This allows each ray to begin at an arbitrary point in the box and with an arbitrary background luminosity (L_{bkg}). The rays are then propagated in parallel towards the observer.

The rays are split into small bins along the line-of-sight, and the quantities from nearby SIMBA gas particles are averaged into each bin. The width of these bins is fixed to be smaller than the distance between neighbouring particles in the highest-density regions. Quantities are weighted both by their distance from the ray bin (using a cubic spline kernel and the smoothing length of each particle) and their H I mass.

The frequency-dependent optical depth for each ray is then calculated by integrating along the line of sight:

$$\tau_\nu = \sigma \phi_\nu(v_{\text{los}}, T) \int n_{\text{HI}} ds \quad (1)$$

where σ is the cross-section from the hyperfine 21cm transition, and the shape of the line is modelled as a Voigt function (ϕ_ν), which captures both Doppler and

natural line broadening, and encodes the line-of-sight velocity (v_{los}) and temperature (T) information of the particles along the ray. When calculating the cross section we must assume a spin temperature, which encodes the distribution of hydrogen atoms across the energy levels involved in the 21 cm spin-flip transition (B. T. Draine 2011). We fix this to $T_s = 300$ K, the value for the Milky Way, a choice discussed in more detail below.

The frequency-dependent luminosity is then given by attenuating the background luminosity from the jet particle by the optical depth:

$$L(\nu) = L_{\text{bkg}} e^{-\tau_\nu} \quad (2)$$

to give the shape of the absorption line for each ray.

When calculating the broadening of the line profile, we use the gas particles' kinetic temperature (T_k) as given by SIMBA. An alternative, observationally motivated choice might be to use a spin temperature (T_s). However, T_s is not provided in the SIMBA particle data catalogues, and it likely varies with redshift and galaxy properties (J. R. Allison 2021). Furthermore, we found that using an assumed, Milky Way like spin temperature ($T_s = 300$ K) produced line widths much narrower than observed, whereas the kinetic temperature ($T_k \approx 10^4$ – 10^5 K for gas particles with high H I) better matched realistic line profiles. This could be because using a fixed T_s neglects any gas turbulence or multiphase H I that is unresolved by SIMBA (sub-kpc scales). However, assuming the kinetic temperature for the H I gas is a good estimate of the combined thermal and turbulent broadening.

2.4. Mock observations

After the ray tracing step, we now have a collection of rays with arbitrary (x, y) coordinates (having been integrated along the z -axis) and H I -attenuated luminosity profiles (L_ν). In this section, we describe how to transform this into an observed radio spectrum. In this paper, we use ASKAP-FLASH as our survey target (J. R. Allison et al. 2022; H. Yoon et al. 2025) and match the observational parameters to those used in the Pilot Survey. However, we note that our pipeline can be applied to any instrument or project that can target the 21 cm line.

Motivated by this survey choice, we fix our galaxies at $z = 0.4$ in this work to match the lower bound of the region in which ASKAP-FLASH is sensitive to H I absorption. The radio luminosity assigned to each ray is then converted into observer units of flux density:

$$S_{\nu, \text{obs}} = \frac{L_\nu}{4\pi D_L^2} (1+z)^{1+\alpha} \quad (3)$$

where L_ν is the radio luminosity of the source (in W Hz^{-1}), $S_{\nu, \text{obs}}$ is the observed flux density (in Jy), D_L is the luminosity distance in parsecs, and α is the spectral index of the source, which we fix at a fiducial value of $\alpha = 0.7$. The $(1+z)^{1+\alpha}$ term combines a relativistic correction and a k -correction for the spectral bandwidth.

The rays are then binned into pixels corresponding to an angular scale of 5 arcseconds (Fig. 1, second column). This pixelated image is then convolved with a 30 arcsecond Gaussian beam to mimic the observation with the ASKAP telescope (third column). As long as the beam is well-sampled ($\gtrsim 6$ pixels/beam) the pixel scale is unimportant. Gaussian noise of $\sigma_{\text{noise}} = 5$ mJy/beam/channel is added to the data to match the per-channel sensitivity of the ASKAP-FLASH survey (H. Yoon et al. 2025).

For many combinations of jet and orientation, this leads to unresolved, point-like sources in mock images at $z = 0.4$, as can be seen in Figure 1. This agrees well with observations, since since the majority FLASH continuum sources are expected to be compact and unresolved in ASKAP images (J. R. Allison et al. 2022).

A spectrum is extracted from the cube towards only the brightest pixel in the image (last column), and with a spectral resolution of 18.5 kHz (corresponding to $\sim 6 \text{ km s}^{-1}$ at $z = 0.4$). Although ASKAP-FLASH spectra use the full 288 MHz bandwidth of the instrument to search the $z = 0.4 - 1$ redshift range, this is not necessary for our mock spectra. Even extreme outflows of 1000 km/s or more would have a width of no more than 2–3 MHz at our fiducial redshift. We therefore restrict the spectrum to a 6 MHz chunk, centred on the bulk velocity of the galaxy. Alongside the spectrum, we generate a channelised ‘noise spectrum’ containing Gaussian noise at the same level as that seen in the ASKAP-FLASH pilot surveys (5 mJy/beam per spectral channel).

The spectrum and associated noise are then fed to the same Bayesian linefinder used on ASKAP-FLASH observations (J. R. Allison et al. 2012). This searches for the presence of a Gaussian absorption line profile anywhere in the spectrum, and once found, it returns the parameters of the best fit model, as well as the likelihood of the line model over a pure noise model, expressed as a Bayes factor. Following the ASKAP-FLASH Pilot Surveys, we consider detections with a Bayes factor $\ln(B) \geq 30$ to be statistically significant (H. Yoon et al. 2025).

From the parameters of the FLASHfinder model, an observed column density of neutral Hydrogen can be derived as follows:

$$N_{\text{HI}} = 1.823 \times 10^{18} \frac{T_s}{f} \int \tau(\nu) d\nu \quad (4)$$

where T_s is the spin temperature of the gas, and f is the covering fraction of the radio source which is subtended by the absorbing gas.

For observers, assumptions need to be made about both T_s and f in order to calculate N_{HI} , which is taken as an estimate of the true column. Although we can directly measure the covering fraction f in our simulations, we too need to make assumptions about the spin temperature, which is not the same as the kinetic temperature of the gas recorded in the SIMBA particle datasets, as already discussed in Section 2.3. In order to compare our simulated H I absorption features to the observations of ASKAP-FLASH as fairly as possible, in this work we therefore rely on direct comparison of the optical depth τ , for which we obtain measures of width ($\Delta\nu$), depth or ‘peak optical depth’ (τ_{peak}) and overall strength or ‘integrated optical depth’ (τ_{int}) directly from the FLASHfinder. Ward, Kerrison & Tonnesen (*in prep.*) will discuss measures of the column N_{HI} in more detail.

3. SAMPLE CONSTRUCTION

Our main motivation is to determine under what conditions H I is detectable given the ASKAP-FLASH survey parameters. Accordingly, we draw our simulated galaxies from a SIMBA snapshot at $z = 0.4$ (snapshot #151), corresponding to the lower bound of the redshift space in which ASKAP-FLASH is sensitive to H I. Each galaxy is cut out from the box using a sphere of 100 kpc radius, and aligned by default so that its average angular momentum vector is vertical and perpendicular to the observed line of sight. This alignment is achieved using the `L.align` function also used by MARTINI.⁵ We outline below additional properties of this sample as relevant to our investigation.

3.1. SIMBA galaxy selection

From the 46,575 galaxies within the SIMBA snapshot, we focus on 266 which are both likely to host a (radio) AGN, and which are H I-rich. In this way, we are testing the *detectability* of H I in ASKAP-FLASH-like mocks. This sample, and our selection process, are described in more detail below.

N. Thomas et al. (2021) defined radio AGN within SIMBA as those galaxies with massive, central black holes ($M_{\text{BH}} > 10^8 M_*$) and full-power kinetic jets ($0 < f_{\text{Edd}} < 0.02$). We follow this prescription, but we relax the requirement on f_{Edd} here to allow for any amount and type of feedback (R. Davé et al. 2019). This choice was made because observationally, ASKAP-

FLASH detections are made towards both quasar-type AGN, and AGN with extremely compact ($\lesssim 1$ kpc) jets, potentially indicative of AGN transitioning between ‘radiative-mode’ and ‘jet-mode’ accretion as implemented in SIMBA.

In summary, our selection criteria were as follows:

1. Galaxy is H I-rich : $M_{\text{HI},30 \text{ kpc}} > 10^{9.5} M_\odot$
2. Galaxy is massive : $M_* > 10^{10.5} M_\odot$
3. Galaxy hosts AGN : $M_{\text{BH}} > 10^8 M_\odot$ and $\dot{M}_{\text{BH}} > 0$

We show our sample in Figure 3, where the shaded box indicates the cuts that were performed to ensure the selected galaxies are both H I-rich (1) and massive (2). This figure also motivates our threshold for H I-rich, since a cut-off of $M_{\text{HI},30 \text{ kpc}} > 10^{9.5} M_\odot$ selects the high-mass mode of the bimodal distribution seen in the AGN kernel density estimate (dotted line, y-axis). The threshold in stellar mass ensures that our galaxies have not undergone any significant mass loss events which would put their stellar mass below the black hole seeding threshold ($M_* \approx 10^{9.5} M_\odot$), but is a more conservative limit than that imposed by N. Thomas et al. (2021). The scattered points coloured by f_{Edd} in Figure 3 indicate those galaxies hosting AGN (3), and the kernel density estimates along each axis show the distribution of each sub-population.

3.2. Sample morphologies

Our full sample of 266 galaxies were visually classified based on their morphology in the $z = 0.4$ SIMBA snapshot. Of these, ~ 30 per cent (107/266) have a regular gas disk structure, ~ 11 per cent (30/266) show spiral structure, and ~ 16 per cent (43/266) have a spherical, bulge-like gas distribution. A further ~ 15 per cent (41/266) are strongly interacting (potentially merging) with a nearby ($d < 100$ kpc) companion, and ~ 11 per cent (31/266) have a disturbed gas profile which may indicate interactions with a more distant companion not visible in our cutouts. We combine these last two categories, of observed and potential interactions into one category of ‘mergers and rogues’ in Section 4.5. A representative subset of the full sample is shown in Figure 4, to give the reader a sense of their diverse (gaseous) morphologies, where it is clear that some (e.g. #446, #1958) have the majority of their H I centrally concentrated within $r \lesssim 20$ kpc, while others show H I structure on larger scales, up to $\lesssim 100$ kpc.

How does this compare to observational samples? Optical imaging of ASKAP-FLASH sources reveals that 18% of the Pilot Survey H I detections have at least one companion within a projected distance of 20 kpc on sky

⁵ Found at https://github.com/kyleaoman/kyleaoman_utilities

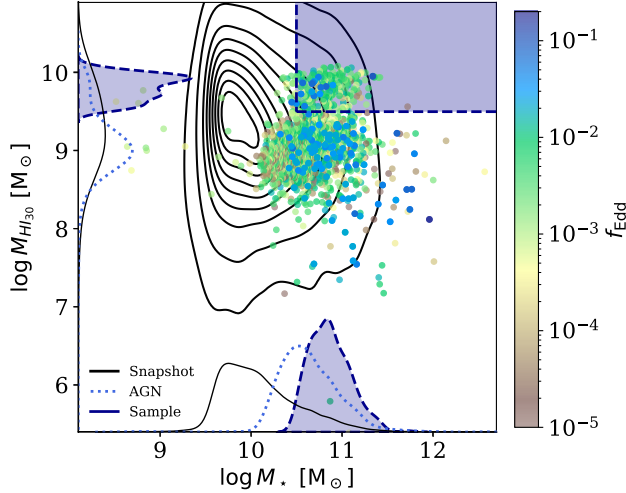


Figure 3. The parameter space used for selecting our SIMBA galaxies. We selected a sample of HI-rich galaxies with moderate to high stellar masses, and which span a representative range of f_{Edd} values. Contours are all galaxies within the SIMBA snapshot, scattered points are those which are considered AGN using the criteria outlined in Section 3.1, and the shaded box indicates the selection criteria used for our sample. Along the x - and y -axes are kernel density estimates showing the distribution of each sub population.

at the redshift of the H I feature (Sadler et al. *in prep*). In addition to this, some targeted searches from the literature seem to suggest that mergers comprise a significant fraction of HI-detected sources. R. Dutta et al. (2018) showed that in a sample of mergers, the detection rate is much higher than in other samples (83 ± 17 per cent), and that the absorption lines themselves are much stronger in these systems. By contrast, the optically-selected sample of K. Geréb et al. (2015) contained only 4/32 mergers, with a further 7/32 with extended optical morphology, and the remainder appearing compact. It is likely that optical selection criteria used for many previous H I searches will influence these fractions, for example by missing strongly dust-obscured companions (I. Langan et al. 2024). Furthermore, the fraction of mergers in an H I-selected sample is not a fair comparison to this set of simulated galaxies because we have not yet determined which of these we would see in H I absorption, we can only say that we ought at least to include *some* mergers in our selection based on these observational results. Amongst the more general population of powerful (jetted) radio galaxies, M. Chiaberge et al. (2015) show that a very high fraction (90 per cent) are associated with recent or ongoing merger activity. The fraction of non-breakout (PS) radio AGN which show evidence of recent mergers or interactions is somewhat more modest, at 40 – 70 per cent (C. P. O’Dea et al. 1996; K. E. Ran-

dall et al. 2011). In all cases, observers tend to classify these galaxies as ‘interacting’ if they show evidence of streams or tidal tails (even without a clear identification of the interacting companion), so a fairer comparison to our sample might be the ~ 30 per cent which we classify as mergers and rogues, as introduced above and discussed further in Section 4.5.

In summary, our sample shows a range of gas morphologies indicative of both isolated galaxies and merging/interacting systems that broadly reflects the diversity of galaxies hosting radio AGN, and detected in H I absorption.

3.3. Radio jet luminosity & morphology choices

As mentioned in Section 2.2, for each galaxy drawn from the SIMBA box we inject radio jets for which we have a number of free parameters including morphology, length and total luminosity. The luminosity of radio AGN is well characterised in the local Universe (T. Mauch & E. M. Sadler 2007) and even out as far as $z \sim 1$ (P. N. Best et al. 2014; M. J. Hardcastle et al. 2025), the upper limit at which ASKAP-FLASH is sensitive to H I. In all cases there is an exponential decay in the high luminosity tail of radio power, so we use these observational constraints to place an upper limit of $L_{1.4\text{GHz}} = 10^{28} \text{ W Hz}^{-1}$ on the integrated luminosity of our radio jets. Although the radio luminosity function shows that there exists an abundance of AGN with jet luminosities $L_{1.4\text{GHz}} \leq 10^{24} \text{ W Hz}^{-1}$, ASKAP-FLASH is only expected to be sensitive to H I towards the most powerful radio galaxies (J. R. Allison et al. 2022). Our pipeline independently confirms that, with ASKAP-FLASH survey parameters, we are not sensitive to H I in absorption with jets fainter than $L \sim 1 \times 10^{24} \text{ W Hz}^{-1}$, so this was set as the floor in the integrated luminosity of our jet models for the remainder of this work.

As discussed in Section 2.2, with SANGRIA we allow for the injection of any arbitrary jet morphology and luminosity into our SIMBA galaxies. This is in contrast to N. Thomas et al. (2021) who explicitly link the radio luminosity of their AGN to the BH properties contained in the SIMBA catalogues. Our reasoning for decoupling the jets from the BH properties in this work is twofold; first, due to boxsize limitations the fiducial SIMBA run does not contain more than a handful of AGN with $L_{1.4\text{GHz}} \geq 10^{27} \text{ W Hz}^{-1}$ using the N. Thomas et al. (2021) prescription. Since these powerful jets are preferentially where ASKAP-FLASH detects H I (recalling that it is a wide and shallow survey), using BH-linked radio luminosities would lead to a prohibitively small sample of FLASH-like H I detec-

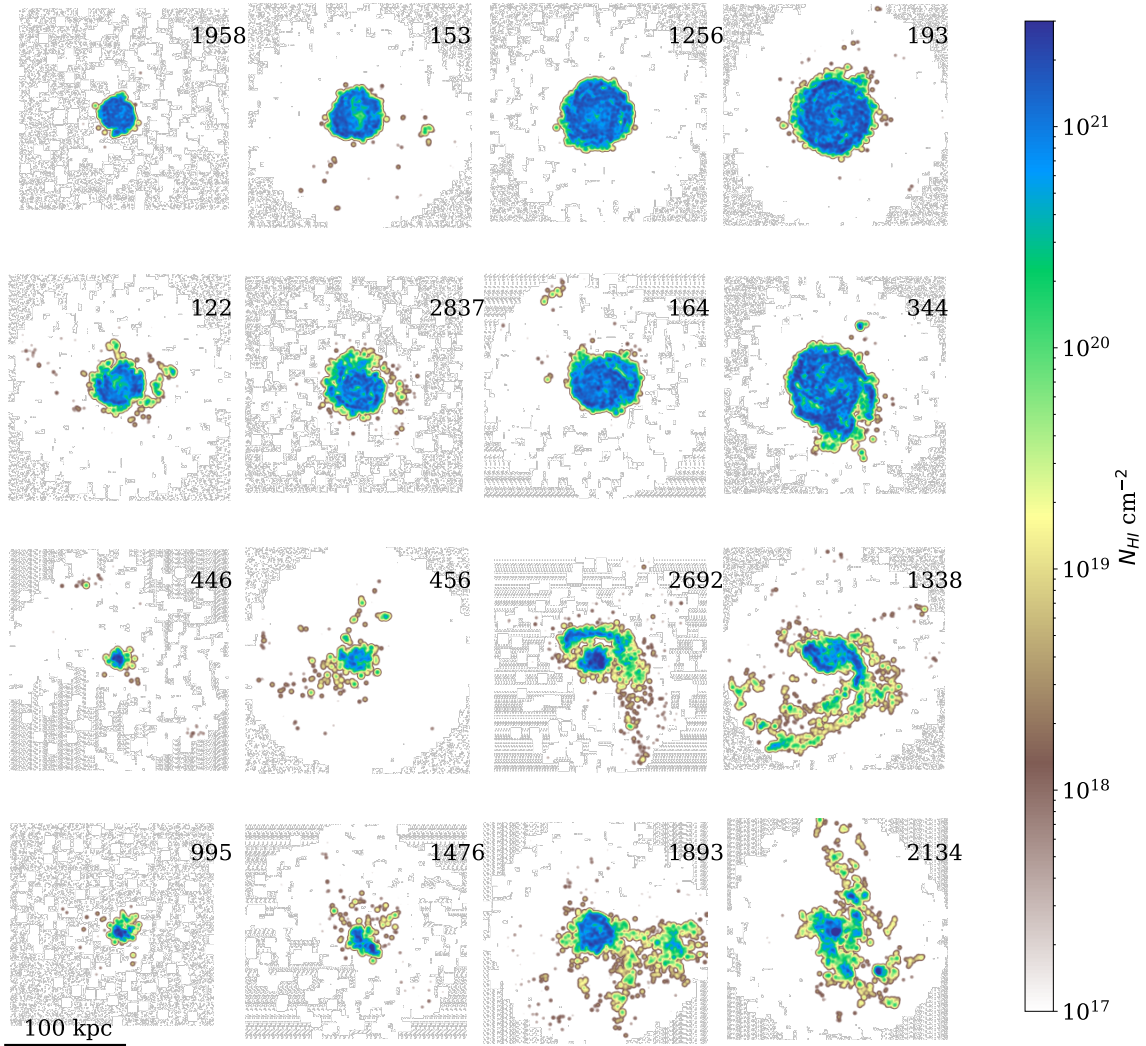


Figure 4. A selection of galaxies from our full sample, showcasing the range of galaxy morphologies. Ordered approximately from small to large (left to right) and from isolated disks to interacting systems (top to bottom) The colorscale indicates the column of H I. All galaxies are oriented such that their angular momentum vector (averaged over all star and gas particles) is oriented out of the page, such that disks are seen face-on. Note that galaxy 164 is often referred to in this paper as an example.

tions. Secondly, as stated elsewhere throughout this paper, by decoupling the jet morphology and luminosity from the BH (and galaxy) properties, we can perform a test of geometric affects in isolation. For the remainder of this work we thus restrict the injected radio luminosities to $10^{24} \leq L \leq 10^{28} \text{ W Hz}^{-1}$, over which range we will perform a uniform parameter study in Section 4, before more closely matching observed distributions in Section 5.

A summary of the different combinations of jet and galaxy parameters used in the remainder of this work is given in Table 1. Thus, in the following section when we refer to the “full sample” we are referring to all jet types, radii, and luminosity combinations for every galaxy. The ‘Gal.’ sample corresponds to those mocks

where the jet and galaxy disk are fixed perpendicular to each other, and rotated through 12 viewing angles (the first row in Table 1). The ‘Jet’ sample is those mocks where the galaxy is fixed at 60 degrees to the line of sight, and the jet is rotated through 12 viewing angles (the second row in Table 1).

4. TESTING DETECTABILITY IN MOCK OBSERVATIONS

With our parameters established in previous sections, we here perform a uniform parameter sweep to explore the conditions under which H I is detected in absorption. In particular, we focus on whether the jet type and luminosity, or the geometrical orientation of the jet and galaxy effect whether ASKAP-FLASH would detect an absorption line (4.1-4.4). We then consider the effect of

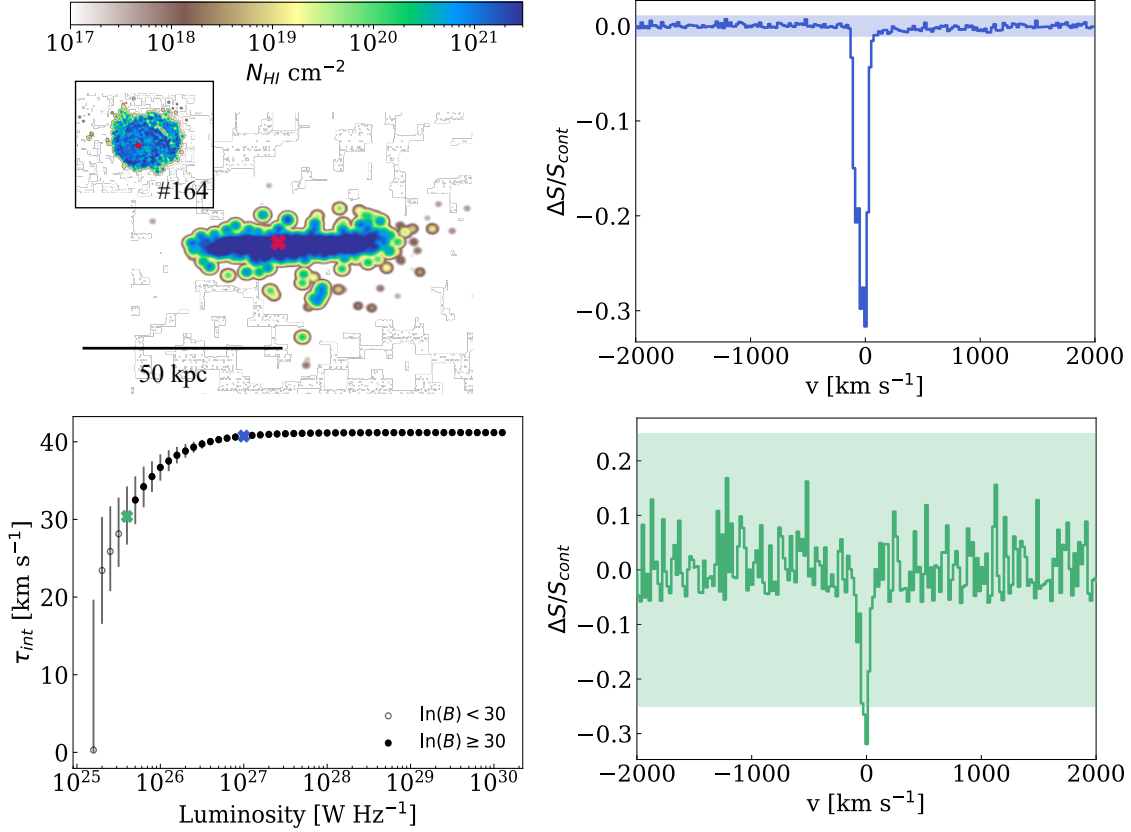


Figure 5. *Left, top:* The arrangement of SIMBA galaxy #164 used to test the effect of radio jet luminosity, aligned with the disk perpendicular to line of sight with an injected quasar (point source) of radio emission at the centre (red cross). *Left, bottom:* The integrated optical depth (i.e. strength) of the H I absorption feature as a function of radio luminosity, detected for this setup using the FLASHfinder. Filled circles indicate a significant detection ($\ln(B) \geq 30$). Two points are marked by crosses for which spectra have been extracted. *Right:* Two mock H I spectra extracted from the setup at a luminosity $L_{1.4} = 1.00 \times 10^{27} \text{ W Hz}^{-1}$ (top), and $L_{1.4} = 3.98 \times 10^{25} \text{ W Hz}^{-1}$ (bottom). The shaded region in each of these plots indicates $5 \times$ the root mean square noise across the spectrum.

Rotation type	Jet Type	Jet Length [kpc]
Galaxy rotation (Gal.)	FR0	0.5
(jet fixed $\perp$ to galactic disk)	FRI	1.0, 5.0, 10.0
	FRII	5.0, 10.0, 50.0
Jet rotation (Jet)		
(galaxy inclined 60° to LoS)	FRI	1.0, 5.0
	FRII	5.0, 10.0, 50.0

Table 1. A summary of the jet-galaxy parameters used in this work. Each of these jet type/radius combinations is modelled with 17 luminosities regularly spaced in log between $L_{1.4\text{GHz}} = 10^{24} \text{ W Hz}^{-1}$ and $L_{1.4\text{GHz}} = 10^{28} \text{ W Hz}^{-1}$.

galaxy morphology (4.5), before ending this section with a discussion of overall detection fractions (4.6).

4.1. The effect of jet luminosity

As part of our jet injection, we specify their integrated radio luminosity at 1.4 GHz in the source rest-frame ($L_{1.4}$), decoupled from the black hole mass and accretion rate, as discussed in Section 3.3.

To show the effect of luminosity on the detection of H I in absorption, we examine a simple disk galaxy (SIMBA galaxy #164) oriented perpendicular to the line of sight, and with an injected quasar (point-like radio emission) at its centre, as shown in Figure 5. We vary the luminosity of this ‘quasar’ within the range $L_{1.4\text{GHz}} \sim 10^{24} - 10^{30} \text{ W Hz}^{-1}$ (larger than the range of our “full sample” to determine the thresholds for non-detection and saturation), creating a mock FLASH spectrum for each luminosity step, which is then fed to FLASHfinder to identify absorption signatures. We see that, although FLASHfinder detects absorption at just a few $\times 10^{25} \text{ W Hz}^{-1}$ (where detection here means sim-

ply that $\tau_{int} > 0$), the integrated optical depth τ_{int} of the detected line is low with a high degree of uncertainty and the Bayes factor does not reach the threshold for significance ($\ln(B) \geq 30$). For reference, we add two crosses to this plot over luminosities for which we show the extracted spectra on the right of Figure 5. The first, shown in the lower right panel, is at the lowest luminosity for which the detection of H I would be considered significant by the FLASH team, with a Bayes factor $\ln(B) \geq 30$, corresponding to an injected $L_{1.4\text{GHz}} = 3.98 \times 10^{25} \text{ W Hz}^{-1}$ ‘quasar’. The detected features are evidently noisy in this spectrum, given the large uncertainty on τ_{int} , although it contains the same structures at approximately the same depth as the spectrum towards the blazing $L_{1.4\text{GHz}} = 1 \times 10^{27} \text{ W Hz}^{-1}$ ‘quasar’, with a Bayes factor two orders of magnitude larger (Figure 5, upper right). It is only at this second, higher luminosity where the optical depth approaches the limiting value of 40 km s^{-1} . This limiting optical depth corresponds to the H I column measured from the particle data if we fix the spin temperature to $T_s = 324 \text{ (K)}$ in our conversion to N_{HI} , though this is slightly higher than the $T_s = 300 \text{ K}$ input into SANGRIA to determine the 21 cm cross section. This may indicate that the Gaussian fit by FLASHfinder fails to recover some complexities in the visible H I profile, or that there is some very low optical depth gas that is strongly Doppler-shifted away from the main feature in the spectrum. A more in-depth discussion of these differences is left to Ward, Kerrison & Tonnesen (*in prep.*). The key takeaway from this investigation though, is that while some FLASH detections with $\ln(B) < 30$ are undoubtedly real, they may fail to recover the full line profile, and hence the full H I column, until a much higher Bayes factor, corresponding to a much higher signal to noise.

Observationally, J. N. H. S. Aditya et al. (2024) found in a flux-limited subsample of ASKAP-FLASH pilot data that the only detections of H I were made against intrinsically luminous radio sources, with $L_{1.4\text{GHz}} \geq 10^{27} \text{ W Hz}^{-1}$. We find significant detections of H I at much lower luminosities than that in this test ($L_{1.4\text{GHz}} \sim 10^{25.5} \text{ W Hz}^{-1}$). We interpret this as an effect of radio morphology; the radio luminosity in these mocks is concentrated in a point-like region at the centre of a high-density gas disk, whereas many radio jets have some spatial extension (more on this below). Overall though, our detection of H I as a function of radio luminosity follows the same trend as seen in ASKAP-FLASH observations, namely that brighter radio jets produce stronger H I features. Physically, this is explained by the fact that radio photons are not sufficiently energetic to photo-ionise ambient neutral gas, so we expect in-

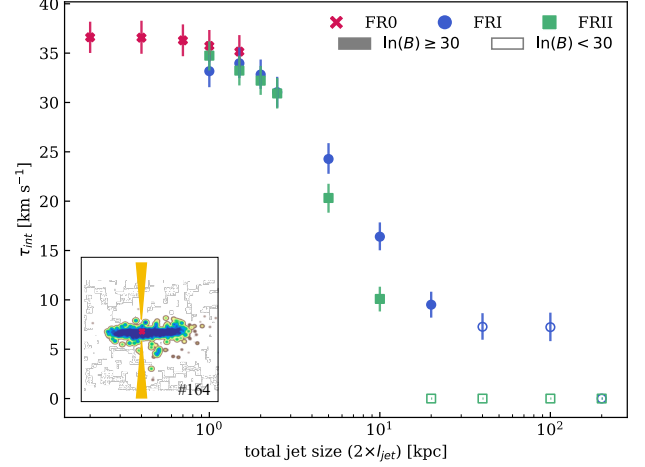


Figure 6. The effect of jet length on the detectability of H I in absorption. The most compact jets are modelled with an unresolved, FR0 morphology (pink crosses), but beyond 1 kpc we insert both FRI (blue circles) and FRII jets (green squares) into this setup. Non-detections ($\ln(B) < 30$) have unfilled markers. We fix the luminosity at $L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$.

creasing radio luminosity to monotonically increase the signal-to-noise of any spectral line detected in H I absorption.

4.2. The effect of jet morphology

Moving from simple, quasar-like radio emission to more complex conditions, we now consider the effects of jet morphology on the detectability of H I absorption. These effects are twofold, due to both the jet length and typology, which we consider separately below.

4.2.1. Jet length

Radio AGN jets have been observed with lengths spanning more than six orders of magnitude in scale, from a few parsecs to more than a megaparsec. However, ASKAP-FLASH is preferentially detecting H I absorption against jets no more than a kiloparsec end-to-end (E. F. Kerrison et al. 2024; H. Yoon et al. 2025). To begin exploring this computationally, we again examine galaxy #164 from SIMBA at $z = 0.4$, this time injecting jets of varying lengths.

We fix the luminosity at $L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$, corresponding to a Bayes factor of $\ln(B) \sim 300$ for point-like, quasar emission in Section 4.1. The jet length is varied across two orders of magnitude, from an unresolved FR0 ($2 \times l_{\text{jet}} = 0.2 \text{ kpc}$) to an extended FRI/FRII ($2 \times l_{\text{jet}} = 200 \text{ kpc}$) that spans the full width of the simulation cutout (which has a radius $r = 100 \text{ kpc}$). For this setup, we again align the galaxy disk edge-on to our line of sight, where the orientation of the disk is identified

using the averaged angular momentum vector of gas and star particles. We then align the jets perpendicular to the galactic disk. The setup is shown as an inset in Figure 6.

As shown in Figure 6, when the jets remain confined to within the disk ($l_{\text{jet}} \lesssim 2 \text{ kpc}$) the integrated optical depth of H I absorption (a proxy for the H I column density) remains constant to within statistical uncertainties though slightly lower than the point source optical depth shown in Figure 5. Once the jets begin to break out of the disk, the integrated optical depth drops exponentially until it falls below the significance threshold for the FLASH survey ($\ln(B) = 30$) at a total length $2 \times l_{\text{jet}} \geq 20 \text{ kpc}$ for an FR II, or $2 \times l_{\text{jet}} \geq 40 \text{ kpc}$ for an FRI. This trend reflects the radial profile of the gas in galaxy #164, where larger jets subtend more of the low density, circumgalactic medium. This is essentially an extension of the radial profiles mapped observationally with intervening Mg II absorbers (R. Bordoloi et al. 2011), and in cosmological mocks of Ly α absorption (S. Weng et al. 2024) down to the lowest impact parameters, and to radio wavelengths. If we perform this same exercise with the disk oriented face-on to our line of sight we find the same trend, but none of the Bayes factors are above 30, and $\tau_{\text{int}} > 0$ is only found for a few jets where $2 \times l_{\text{jet}} < 1 \text{ kpc}$.

For this test we have been careful to describe total jet lengths ($2 \times l_{\text{jet}}$), to show that our threshold for jet breakout corresponds to that seen observationally in PS sources ($\sim$ a few kpc), but for the remainder of this work we will discuss our results directly as a function of l_{jet} .

4.2.2. The jet type - N_{HI} connection

Associated H I absorption has been detected towards both FRI and FR II-type AGN, both in front of the compact radio core (R. Morganti et al. 2001), and as jet-driven outflows (E. K. Mahony et al. 2013), but the impact of morphological differences on detection rate is difficult to pinpoint observationally. As seen already in Figure 6, there is a difference between the strength of H I absorption detected towards an FRI and an FR II once the jet length extends beyond the disk. For the same total luminosity and size, the FR II-type jets produce systematically weaker absorption signatures. In the setup presented in Section 4.2.1, the discrepancy in τ_{int} at a jet length 2.55 kpc is $4 \pm 3 \text{ km s}^{-1}$, this extends to $6 \pm 3 \text{ km s}^{-1}$ at a jet length of 5 kpc. To ensure this is not simply a product of the galaxy chosen, we extend this analysis to our Gal. sample of 266 galaxies in the ‘Galaxy rotation’ setup, but we keep the angle fixed such that line of sight is into the plane of the disk. The results of this analysis are shown in Figure 7, where we present a

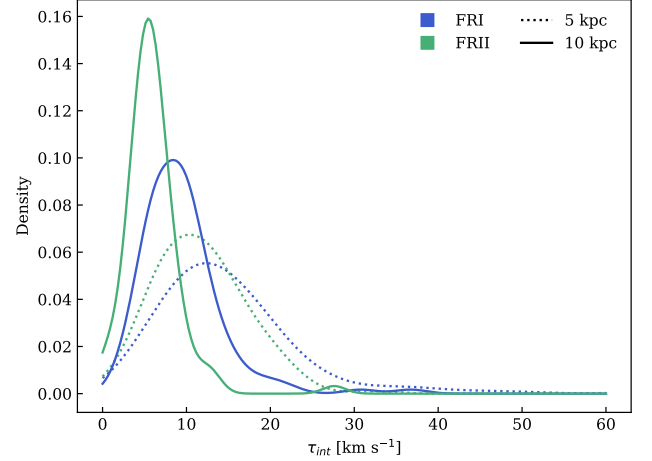


Figure 7. The distribution in integrated optical depth (τ_{int}) across all 266 galaxies in our sample when observed with an FRI (blue) and FR II type jet (green). We show the results for both 5 kpc (dotted) and 10 kpc (solid) jets with an integrated luminosity $L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$. The distributions are plotted using a gaussian kernel density estimation (KDE).

kernel density estimate (KDE) for the integrated optical depth across this sample, broken down into detections made against FRI/FR II jets (line colour), each at 5 and 10 kpc lengths (linestyle).

It is clear that though modest, this difference in τ_{int} persists across the whole sample. At $l_{\text{jet}} = 5 \text{ kpc}$ detections against FRI-type jets have a median optical depth 1.86 km s^{-1} larger than those towards FR II-types (comparing the dotted lines), and this difference extends to 3.1 km s^{-1} in $l_{\text{jet}} = 10 \text{ kpc}$ jets (comparing the solid lines). This must therefore be a product of the different luminosity distributions in the jets, with core-dominated FRIs preferentially illuminating the dense gas of the inner galactic regions. We note here that all jets below $l_{\text{jet}} = 100 \text{ kpc}$ are unresolved by the ASKAP beam, but from this figure it is clear that the jet length and morphology make a significant difference to what is happening on ‘sub-beam’ scales, which manifests as differences in the observed H I line profile.

4.3. The effect of galaxy orientation

So far our experiments have maintained a fixed line of sight through the galactic disk(s). However one of the advantages of mock observations is that this need not be the case, and we are free to choose whatever viewing angle we please. We consider here the effect of viewing angle on our ability to detect H I absorption.

4.3.1. A single galaxy

In Figure 8, we show the strength of H I absorption observed towards galaxy #164 illuminated with a

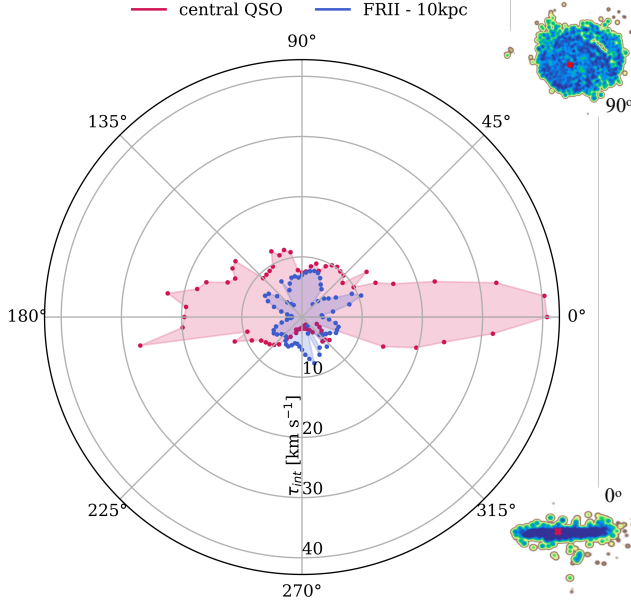


Figure 8. The integrated optical depth τ_{int} as a function of inclination angle of the galaxy, where 0° corresponds to the disk edge-on, 90° to face on. Red indicates the rotational absorption profile seen towards a central ‘quasar’ (pointlike emission), while the blue shaded region is that of an extended FRII. We fix the luminosity at $L_{1.4\text{GHz}}=10^{27} \text{ W Hz}^{-1}$.

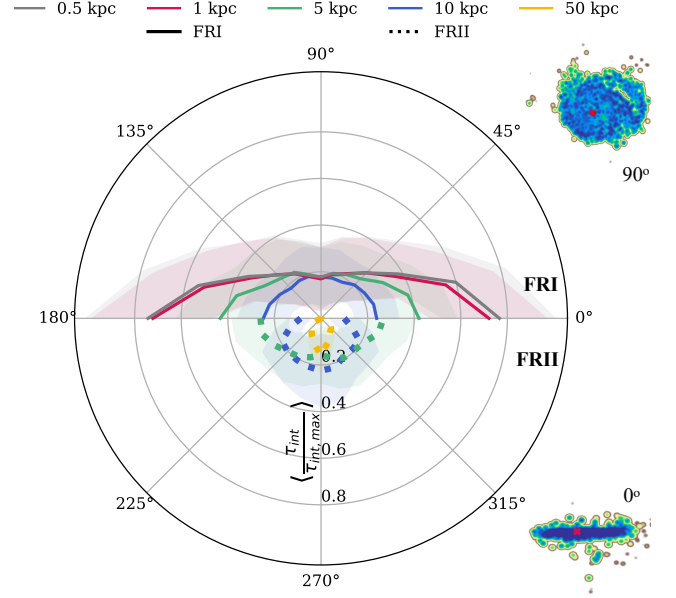


Figure 9. Similar to Figure 8 but for the full sample and all extended jet types. The top half of the plot is for FRI-type jets, the bottom half for FRII-type. Measured is the ‘normalised optical depth strength’, $\langle \frac{\tau_{int}}{\tau_{int,max}} \rangle$, calculated as the optical depth for each angle, normalised by the maximum optical depth attained for that galaxy. Lines are the mean across the full sample. As in Figure 8, 0° corresponds to the disk edge-on, 90° to face on. We fix the luminosity at $L_{1.4\text{GHz}}=10^{27} \text{ W Hz}^{-1}$.

$L_{1.4\text{GHz}}=10^{27} \text{ W Hz}^{-1}$ quasar and rotated through 360° (red shaded region) in 5° steps (note this is a much finer sampling of angles than that used for our full Gal. and Jet sets of mocks). The strongest H I features are seen when the disk is edge-on to our line of sight (0° and 180°), reflecting the fact that these sightlines trace the highest column of gas. We note that even though both edge-on views show more absorption than those closer to face-on, the τ_{int} values still differ by about a factor of 2 between 0° and 180° . Thus the galaxy we have chosen here, although classified as a well-ordered disk, shows strong evidence of disk asymmetry in its H I absorption profile. This phenomenon is well documented observationally (C. J. Jog & F. Combes 2009), and in other cosmological simulations (E. L. Lokas 2022).

If the jets extend far beyond the H I disk, an edge-on inclination does not show significantly stronger absorption features than any other inclination angle. This is represented by the blue shaded region in Figure 8, which describes the H I profile seen towards an FRII with $l_{jet} = 10 \text{ kpc}$. In this case, the strength of the H I feature shows a much weaker dependence on the inclination angle of the galactic disk, and it does not reach an optical depth greater than $\tau_{int} = 10.5 \text{ km s}^{-1}$ for any inclination. At this luminosity, these detections are statistically significant, but towards fainter jets this would quickly become a non-significant detection based on the

results from Section 4.1. Of course, this is but one of our 266 galaxies, we consider next the results across the full galaxy rotation (‘Gal.’) sample.

4.3.2. The entire galaxy population

Across the full Gal. sample the trend is much the same as in the case of Galaxy #164, namely that galaxy inclination has the strongest effect on the absorption profile when the jets are fully embedded within the gas disk ($l_{jet} \leq 1 \text{ kpc}$). This is shown in Figure 9, where the top half of the figure shows the angular profile of all FRI-type jets in our simulations, and the bottom half is reserved for FRII-type jets. Since we want to see how the H I profile changes statistically across a variety of galaxies with different gas masses and distributions, we have first normalised the H I absorption strength (τ_{int}) across each galaxy by the maximum optical depth obtained for any jet configuration within that galaxy ($\tau_{int,max}$) to produce what we call a ‘normalised optical depth strength’ ($\langle \frac{\tau_{int}}{\tau_{int,max}} \rangle$). As we have shown in Figures 6 & 8, a quasar viewed edge-on will often result in the most absorption. This, in combination with the asymmetry of most of our galaxies (e.g. Figure 8), is why

even our smallest FRI jets do not often have $\frac{\tau_{int}}{\tau_{int,max}}$ values.

Figure 9 reinforces the impact of jet morphology as discussed above in Section 4.2.2, where the solid (dotted) lines are the mean values of the sample for FRI (FRII) jets of each length, and the shaded regions denote the 1σ standard deviation across the sample. Core-brightened FRIs produce deeper absorption line profiles when the galaxy is close to edge-on, even at the same overall jet length, than do their edge-brightened counterparts (FRIIs). At the extreme, the 50 kpc FRIIs show much weaker lines across the Gal. sample than any other jet type, since so little of the integrated luminosity is in a region where it backlights dense H I. This means that even for extremely H I-rich galaxies like those in our sample, jet morphology plays a role in the detectability of H I absorption across a range of galactic viewing angles, and the effect is most exaggerated in edge-on disks. As the disk approaches a face-on viewing angle, the impact of morphology shrinks to be almost negligible, since the half of the jet furthest from the observer will always illuminate the galactic gas disk with the same integrated luminosity, just distributed slightly differently due to jet morphology.

4.4. The effect of jet orientation

A galaxy need not always have radio jets aligned perpendicular to the galactic disk. Observationally, X. Zheng et al. (2024) found that the degree of jet-galaxy alignment was dependent on galaxy properties, and S. L. Jung et al. (2025) showed that jet-galaxy alignment is more randomised close to cosmic filaments, with the jets preferentially aligned along the filament rather than the galaxy minor axis (likely a product of merger activity). Moreover, D. Mukherjee et al. (2018) demonstrated in idealised simulations that sub-kiloparsec scale jets initially aligned close to the plane of the galaxy could still drive kiloparsec-scale outflows akin to those seen observationally in multiphase gas. We now consider cases where our injected jets are not aligned perpendicular to the gas disk of the galaxy, and are instead allowed to vary.

In performing the following analysis, we fix the orientation of our disks to an inclination angle of 60° (where 0° is edge-on to the observer, as in Section 4.3), as to first order this is the median inclination angle for a random, uniform distribution of galaxy orientations in three dimensions (it is also the approximate peak of the distribution of SDSS galaxies studied in L. Morselli et al. 2016).

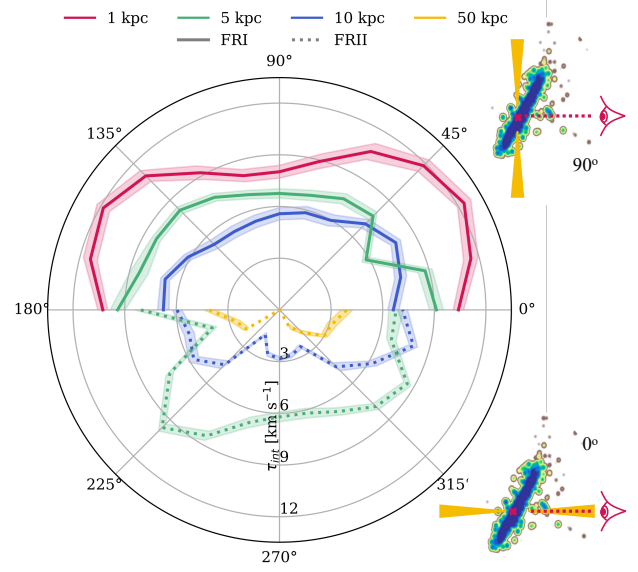


Figure 10. The integrated optical depth τ_{int} for Galaxy #164 as a function of inclination angle of the jet, where 0° corresponds to the jet viewed ‘down the barrel’, 90° to the jet projected across the field of view. Shaded regions around each line indicate the uncertainty on τ_{int} , as reported by the FLASHfinder. We fix the luminosity at $L_{1.4\text{GHz}}=10^{27} \text{ W Hz}^{-1}$.

4.4.1. A single galaxy

As with galaxy orientation in Section 4.3, we consider first the effect of jet orientation on a single galaxy, taking SIMBA Galaxy #164 as our test case. This time, we test all jet lengths used in our full suite of mock observations, shown in Figure 10, again fixing the jet luminosity to $L_{1.4\text{GHz}}=10^{27} \text{ W Hz}^{-1}$ to ensure a high signal to noise on all possible detection configurations.

To orient ourselves, we can first compare the τ_{int} values in Figure 10 to those in Figure 8. Because the galaxy is inclined far from edge-on with respect to our line of sight, the τ_{int} values are much lower than the maximum values in Figure 8. In addition, rotating the jet while keeping the galaxy fixed leads to much smaller variations in the τ_{int} values than rotating the galaxy and jet in the FRI and smaller FRII jets, while the larger 10 kpc FRII jet actually varies by a factor of two depending on its orientation. This is likely because changing the orientation of a longer jet means that very different column density gas in the disk will be the brightest backlit. As with our conclusions about jet type and galaxy orientation, the amount gas between the illuminating source and the observer is the key factor driving variations in τ_{int} . In this setup, the FRI and FRII jets of the same length (same colour in the plot) produce different H I profiles even at inclinations of 0° and 180° , when the jets are viewed down the barrel. This is a result

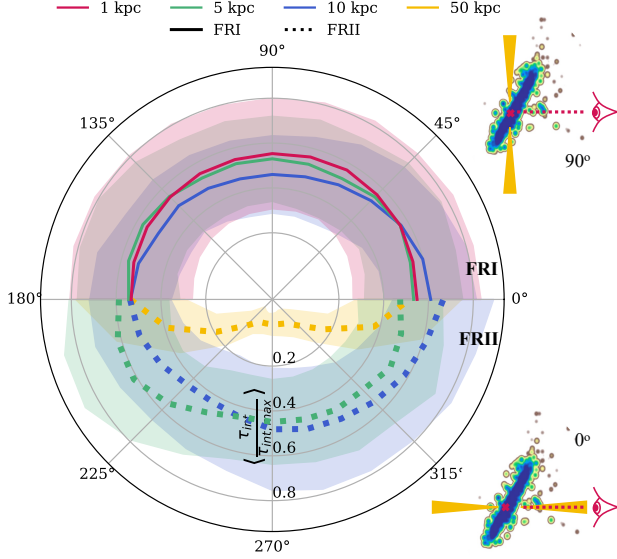


Figure 11. Similar to Figure 10 but for the full sample. This time, lines are the mean across the full galaxy sample, and shaded regions are $1 - \sigma$ standard deviation. The y -axis is also the relative H I absorption strength per galaxy, calculated as the strength of any given H I detection for that galaxy, divided by the strongest measured H I line. We fix the luminosity at $L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$.

of the luminosity being concentrated within a smaller cross-section in core-dominant FRIs compared to edge-brightened FRIIs, which in galaxy #164 reveals variations in H I across the disk, and is an effect we would expect to wash out across all 266 galaxies.

We also note that there are asymmetries in the H I strength over the range of inclinations tested here, especially towards compact jets (FRI, $l_{\text{jet}} = 1 \text{ kpc}$, red solid line in Figure 10). We interpret this again as a signature of the gas disk morphology in galaxy #164, since these asymmetries wash out as the jets extend in length and the integrated luminosity backlights a progressively larger area of the disk. Indeed at a jet length of 50 kpc (yellow dotted line) the original asymmetry is gone and we see instead that the strongest H I absorption is down the barrel of the jet (0°) in a configuration where the entire back half of the jet illuminates the galaxy. If the initial asymmetry seen for compact jets towards Galaxy #164 is a unique product of the H I distribution in this galaxy, we also would not expect to see any such periodicity across the full galaxy sample. This is investigated below.

4.4.2. The entire galaxy population

When considering the effect of jet inclination angle across the full galaxy sample as seen in Figure 11, the asymmetry at small jet lengths is not apparent. This is despite the fact that all galaxies are inclined identi-

cally at 60° to the line of sight. Indeed when averaged across the full sample, jet orientation shows only a very minor variation with angle in any FRI-type configuration (top half of Figure 11). Across the FRIIs there is more variation across 180° of jet rotation, but the 5 and 10 kpc configurations agree to within the 1σ spread (shaded regions) shown on the plot. Furthermore, there is little variation in the median of normalised optical depth strength across the range of jet rotations tested. The most pronounced difference appears for the 50 kpc, FRII jets, which show a marked drop in normalised optical depth once the jet moves more than $\sim 20^\circ$ away from a ‘down-the-barrel’ configuration, parallel to the line of sight. Again, this makes sense as a 50 kpc jet viewed at an angle is likely to illuminate low column density gas far from the galaxy centre due to both their length and the large size of the radio lobes.

4.5. The Impact of Galaxy Morphology

While our previous results considered all 266 galaxies in the Gal. and Jet sample at once, here we compare the mock observations of galaxies separated by morphology. In characterising our sample we have taken care to separate out galaxies with ordered disks from those which might observationally be considered bulge-dominated ellipticals, or morphologically unusual systems caused by recent galaxy-galaxy interactions. Our sample selection ensures all such systems we study are H I-rich, so if we see any differences in our detection fraction across the sample it must be due to the spatial distribution of H I.

Sadler et al. *in prep.* analyse the optical properties of ASKAP-FLASH detections from the Pilot Surveys. They identify a higher fraction of associated H I detections in bulge-dominated systems as opposed to disk-dominated ones, and these detections tend to have higher H I velocity widths ($\Delta\nu$), though these findings are across a small (30-detection) sample. We test here whether similar trends are observed in our high gas mass sample of SIMBA galaxies.

In Figure 12, the left panels present the linewidths of statistically significant ($\ln(B) \geq 30$) H I detections made towards our entire SIMBA sample with the galaxy-jet system rotating (top) and galaxy inclination fixed (bottom), for all jet variations, and a radio luminosity $L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$. The data are divided into galaxies classified morphologically as disks (light histogram), those with more spherical gas distributions, which might observationally be classified as ‘bulge-dominated’ (dark histogram), and ‘mergers and rogues’ which are actively merging, interacting, or have an extended asymmetric distribution setting them apart from narrow disks or spherical bulges (gray histogram).

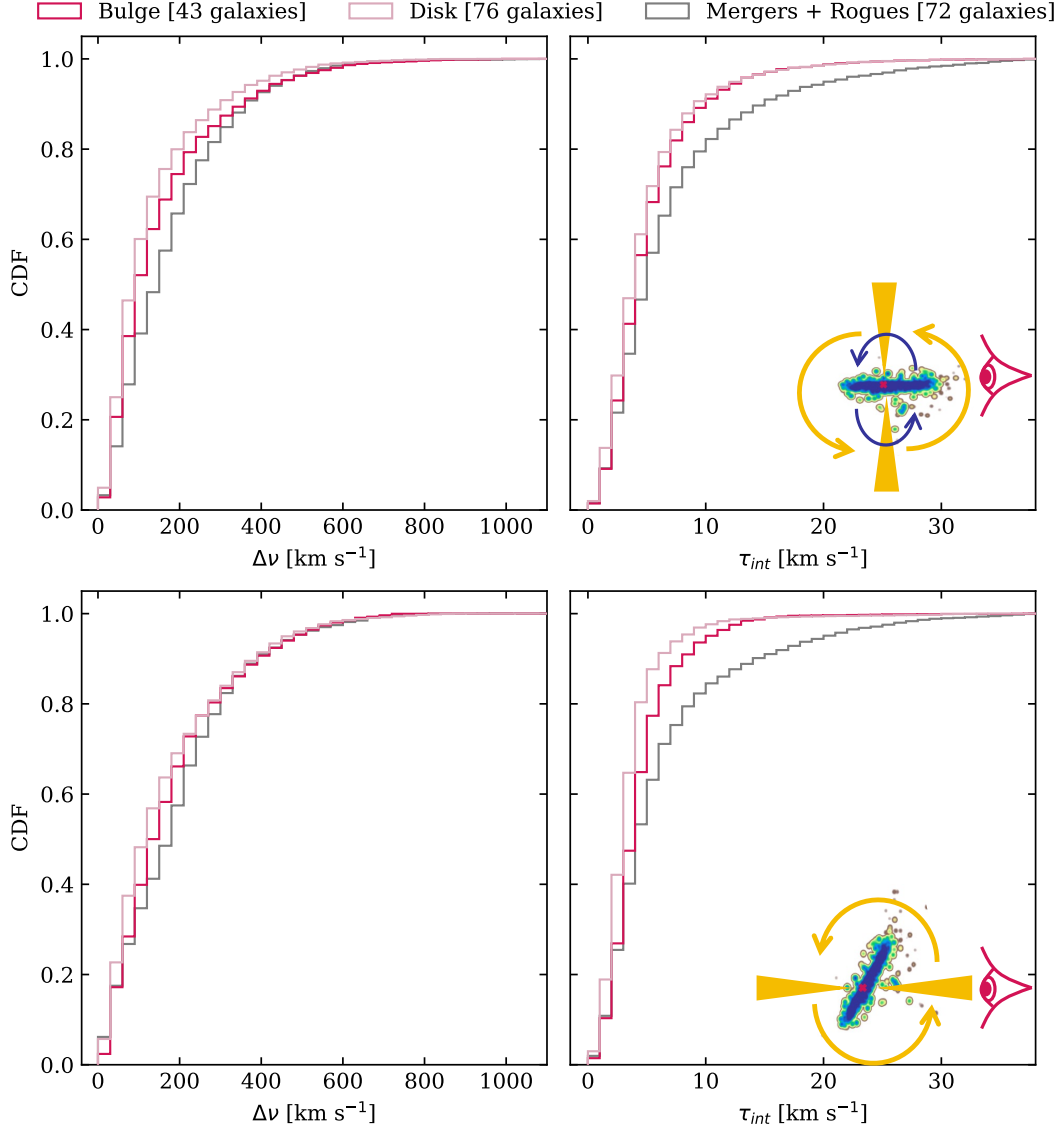


Figure 12. *Left:* The linewidths of statistically significant ($\ln(B) \geq 30$) H I detections made towards the entire SIMBA sample with all jet variations, and a radio luminosity $L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$. *Right:* The same but for the overall H I absorption strength (τ_{int}). *Top:* For the ‘Galaxy rotation’ set of mocks, indicated in the cartoon by both the galaxy (blue arrows) and jet (yellow arrows) rotating together. *Bottom:* for the ‘Jet rotation’ set of mocks, in which only the jet (yellow lines) rotates, while the galaxy remains inclined by 60 degrees to the line of sight..

This final category of ‘mergers and rogues’ most closely resembles the ‘interacting’ category discussed in Section 3.2. As a reminder, 18% of ASKAP-FLASH Pilot Survey detections appear to have nearby ($\lesssim 20 \text{ kpc}$) companions, and previous observational studies have found a high H I detection rate towards mergers (R. Dutta et al. 2018).

We note here that the sub-samples in Figure 12 do not sum to 266, because some galaxies in our sample could not be straightforwardly classified as bulge-dominated, disk-dominated or obvious mergers by eye.

We perform a two-sample Kolmogorov-Smirnov test with the hypothesis that H I widths in disk galaxies are to be equal those in ‘bulge-like’ galaxies. Even assuming a strict threshold of 0.01, we obtain a p -value orders of magnitude less than this for both our ‘Galaxy rotation’ sample and the ‘Jet rotation’ sample. Therefore, we reject the null hypothesis, with our alternative being that H I widths tend to be different in bulge-like SIMBA galaxies compared to disks. In the left-hand panels of Figure 12, the CDF of bulge-dominated galaxies is systematically found to the right of disk-dominated ones, which we further interpret as evidence that the

	ASKAP-FLASH	ASKAP-FLASH	SIMBA Gal.	SIMBA Gal.	SIMBA Jet	SIMBA Jet
	mean $\Delta\nu$	median $\Delta\nu$	mean $\Delta\nu$	median $\Delta\nu$	mean $\Delta\nu$	median $\Delta\nu$
	[km s ⁻¹]	[km s ⁻¹]	[km s ⁻¹]	[km s ⁻¹]	[km s ⁻¹]	[km s ⁻¹]
Bulge-dominated	126.4 ± 14.1	114.6	165.9 ± 2.7	114.8	191.3 ± 2.7	150.0
Disk-dominated	93.2 ± 13.7	97.7	142.1 ± 3.4	96.7	177.8 ± 2.1	125.9
Mergers + rogues	--	--	190.2 ± 2.0	154.7	206.3 ± 2.1	185.0

Table 2. Summary statistics for the width of absorption lines ($\Delta\nu$) across the ASKAP-FLASH Pilot Surveys, and our two SIMBA samples, with the galaxy-jet system rotating uniformly (‘Gal.’) and the jet rotating with respect to a galaxy fixed at a 60° inclination (‘Jet’).

H I widths of bulge-like SIMBA galaxies are larger than those of disks, as was observed in the ASKAP-FLASH Pilot sample. The mergers and rogues also have slightly wider lines again. We also give the mean and median of the linewidth distributions $\Delta\nu$ for the sub-populations in the ASKAP-FLASH Pilot Survey, and in our simulations, in Table 2. These are in line with the results of the KS test.

Performing the same KS analysis on overall H I absorption strength (τ_{int}) rather than width, we once more reject the null hypothesis that the H I line profiles are drawn from the same underlying distribution (p -values again order of magnitude below our cut-off of 0.01 for both the ‘Galaxy rotation’ sample and the ‘Jet rotation’ sample). The CDF for H I absorption strength is given in Figure 12, right, which shows that the mergers and rogues tend have the strongest lines of the three populations, followed by the bulge-dominated galaxies, and then finally the disks.

In performing this analysis, we acknowledge that by eye the CDFs in Figure 12 have similar shapes. However, since each galaxy has been observed across 12 different angles with 7 different jet morphologies, the samples of 43/76/72 galaxies indicated in the figure legend actually translate to 2,961/5,218/4,947 significant H I detections which contribute to the CDF. The observed difference in these distributions therefore statistically significant. Furthermore, each of these categories has a similar rate of non-detections, falling within 11.8 ± 0.1 per cent, so the difference in shape is unlikely to be due to a difference in overall detection rate. We note that this non-detection rate is high even compared to R. Dutta et al. (2018), who had a non-detection rate of ~ 17 per cent. However, we have chosen to fix our jet luminosities high ($L_{1.4\text{GHz}} = 10^{27} \text{ W Hz}^{-1}$), so we caution against over-interpreting this figure here.

Not shown in these CDFs is the upper limit on the integrated optical depth τ_{int} detected in our SIMBA samples. This was done to better highlight the discrepan-

cies between different sub-populations at low τ_{int} . However, we note the strongest H I absorption profile detected to date was towards MRC 0531-237, detected as part of the ASKAP-FLASH Pilot Survey sample, with $\tau_{int} = 143.80 \pm 0.35 \text{ km s}^{-1}$ (J. N. H. S. Aditya et al. 2024). This is near the limit of H I absorption strength detected in our mock observations, bearing in mind that we have designed our sample to contain those galaxies with the highest H I masses from the SIMBA box. Only one SIMBA galaxy (#3055) produces mock line profiles as strong as MRC 0531-237, and it is classed as a merger or rogue. It will be interesting to see just how rare these extremely strong H I profiles are in the full FLASH survey; their occurrence rate may be an interesting test for the distribution of cool, neutral gas in cosmological simulations.

Overall then, our mock observations tell us that compared to disks, we are more likely to detect H I absorption profiles that are both wider and deeper towards galaxies with spherical gas distributions, which might observationally be classified as ‘bulge-dominated’, and profiles that are wider and deeper again towards ‘mergers and rogues’. If merging/interacting galaxies do tend to have wider and stronger H I profiles, it may explain the high detection rate towards these objects in targeted samples (e.g. R. Dutta et al. 2018, , and the discussions in Section 3.2 of this work). The differences between bulges and disks found here are in line with the tentative findings in Sadler et al. *in prep.*, but an analysis of the optical counterparts in the full FLASH survey will be necessary to better characterise this dependence on host-galaxy morphology with greater statistical significance.

4.6. Where can FLASH detect H I ?

To summarise what we have learned so far; ASKAP-FLASH is most sensitive to H I absorption against bright, $L_{1.4\text{GHz}} \geq 10^{27} \text{ W Hz}^{-1}$ jets (Section 4.1), which are compact ($l_{\text{jet}} \lesssim 3 \text{ kpc}$; Section 4.2.1) or at least core-

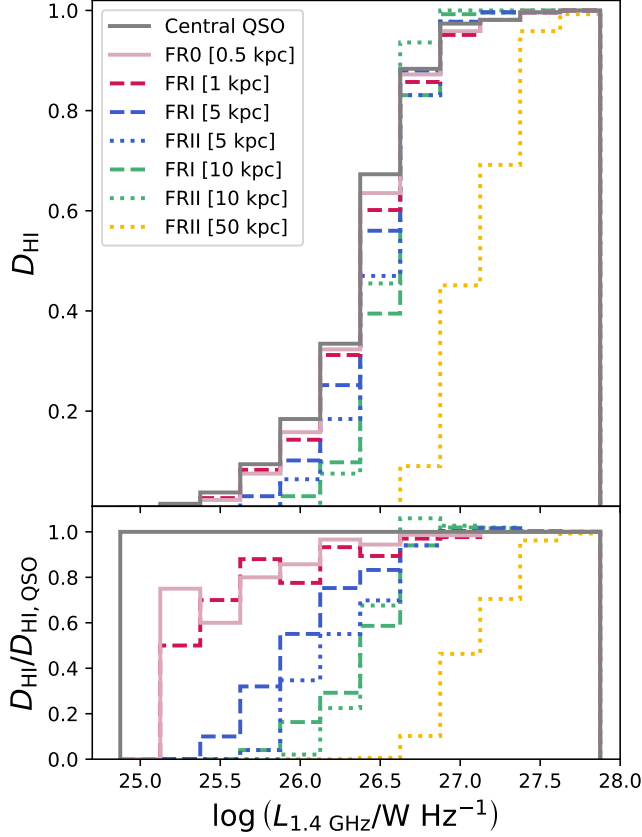


Figure 13. The HI detection fraction as a function of jet luminosity for the ‘Jet Rotation’ setup. *Top panel:* fraction of all galaxies and sightlines that result in a detection of HI, split into the different jet types (solid lines: central quasar/FR0; dashed lines: FRI; dotted lines: FRII) and jet sizes (pale pink: 0.5 kpc; pink: 1 kpc; blue: 5 kpc; green: 10 kpc and yellow: 50 kpc). *Bottom panel:* the HI detection fraction for each jet type relative to the quasar detection fraction. The compact (≤ 1 kpc) jets are detected in HI absorption at a higher rate at low luminosity ($\leq 1 \times 10^{26}$ W Hz $^{-1}$) than more extended jet morphologies.

dominant (FRI; Section 4.2.2). Towards these compact jets, edge-on galaxy disks produce the strongest absorption lines (Section 4.3), though at orientations approaching 90° jet morphology has less of an impact on HI profile depth, and jet inclination has the strongest effect at the largest scales ($l_{\text{jet}} \gtrsim 50$ kpc; Section 4.4). Across our full sample, we tend to see wider and stronger lines towards bulge-dominated galaxies than disk-dominated galaxies, and the widest and strongest lines are seen towards merging or highly-interacting systems (Section 4.5). This last observation is in line with the early results from the ASKAP-FLASH Pilot Surveys (Sadler et al. *in prep.*).

To synthesise some of this information, we now consider the detection *fractions* as a function of jet morphology, recalling that our original motivation for this

investigation was to see whether we could explain the high detection rate towards compact jets in the ASKAP-FLASH sample. In Figure 13, we show the fraction of galaxies and sightlines at a given luminosity with significant HI detections (D_{HI}). We split this into the various jet types and lengths, as denoted by the coloured lines. The lower panel shows the detection fraction relative to the case with a central quasar to more easily distinguish between the different jet morphologies.

We can see from this plot that the probability of detecting HI rises rapidly as a function of luminosity, from $D_{\text{HI}} \lesssim 10$ per cent for $L_{1.4\text{GHz}} \lesssim 10^{25.5}$ W Hz $^{-1}$ to $D_{\text{HI}} \gtrsim 90$ per cent for $L_{1.4\text{GHz}} \gtrsim 10^{27}$ W Hz $^{-1}$. However, within this luminosity range, the size and luminosity distribution of the illuminating radio source plays a large role in determining HI detection. For example, at $L_{1.4\text{GHz}} = 10^{26}$ W Hz $^{-1}$, around $D_{\text{HI}} \simeq 40$ per cent of galaxies with a central quasar are detected in HI. Centrally-concentrated FR0s are slightly less efficient, with the lower panel showing a relative detection of $\simeq 90$ per cent compared to quasars. Extended sources are even less likely to be detected, with FRI-5kpc being detected at rates of 40 per cent compared to quasars; FRII-5kpc at 22 per cent; and the large FRII-10kpc having only a 7 per cent relative detection rate, corresponding to only $D_{\text{HI}} \lesssim 2$ per cent of galaxy sightlines with this jet arrangement having a positive HI detection. These differences decrease at higher radio luminosities, with all jet types being detected at $D_{\text{HI}} = 90$ per cent rates in systems with $L_{1.4\text{GHz}} \gtrsim 10^{26.75}$ W Hz $^{-1}$.

The striking conclusion from this figure is how steeply the probability of detecting HI rises with radio luminosity. Increasing the intrinsic radio luminosity of the source by just over a single order of magnitude ($L_{1.4\text{GHz}} \simeq 10^{25.5-27}$ W Hz $^{-1}$) changes the detection fraction from almost 0 to 100 per cent. This suggests a sharp, low luminosity cut-off for HI detections at around $L_{1.4\text{GHz}} \simeq 10^{26}$ W Hz $^{-1}$, which agrees with the high radio luminosities seen for ASKAP-FLASH Pilot detections (J. N. H. S. Aditya et al. 2024).

5. HI DETECTIONS ACROSS JET POPULATIONS

We have tested the effects of jet morphology, luminosity, and viewing angle in a uniform way across our full sample of HI-rich, AGN-hosting SIMBA galaxies. In order to make meaningful statements about detection rates in ASKAP-FLASH and, crucially, return to the question of over-detection towards compact sources, we need to incorporate a more physically-motivated distribution of radio luminosities and jet types into our suite of mock observations. For the following work, we focus on the ‘Jet rotation’ set of mock observations, where the

galaxy inclination is fixed at 60° to the line of sight, to reflect the ‘average’ galaxy seen in an observational survey, and to allow the jet direction to vary relative to the plane of the galaxy.

5.1. Matching observed populations

We first re-weight our jet luminosities, so that the distribution of jets in our sample matches the observed luminosity function of so-called ‘jet-mode’ radio AGN at $0.5 < z < 1.0$ described in [P. N. Best et al. \(2014\)](#). This is achieved by binning our mock observations by luminosity, and then scaling the bins by a weight, $w_L = N_{\text{target}}/N_{\text{bin}}$, where N_{target} is derived from Table 2 of [P. N. Best et al. \(2014\)](#). In Figure 14 we show this parent distribution of radio luminosities (gray line) and the resulting distribution of H I detections (gray histograms). We also break down the distributions by jet type and length, showing the scaled parent distribution by coloured dotted lines, and the predicted H I detections in coloured histograms, with the top panel containing the luminosity distribution of compact (FR0 and 1 kpc FRI) jets, the middle the broken-out FRI (5 kpc and 10 kpc) jets, and bottom the broken-out FRII (5 kpc, 10 kpc, and 50 kpc) jets. We see that the largest effect is on the high luminosity end of our detections, for which the effective number density is significantly reduced. By contrast, below $L_{1.4\text{GHz}} \sim 10^{26.75} \text{ W Hz}^{-1}$ the luminosity distribution of H I detections remains unchanged, and as we move towards lower luminosities more compact sources tend to dominate. We currently make the assumption that each jet type follows the same distribution, i.e. the ratio between the different jet types is constant as a function of luminosity. A more finely-tuned re-weighting is left for future work, since this would require that the independent luminosity functions of FRIs and FRIIs be well constrained at $z \sim 0.5$. This may soon be possible using future releases of LOW-Frequency ARray (LOFAR) Two-metre Sky Survey (LoTSS; [T. W. Shimwell et al. 2017](#)), and the Evolutionary Map of the Universe (EMU; [R. P. Norris et al. 2011](#)). In the following section we motivate our split between the jet types and show the distribution in Table 3.

Next we must determine our threshold for a simulated jet to be ‘broken out’ of its host galaxy. Observationally, the broadband radio spectral energy distribution (SED) is a proxy for this, if we assume that any radio AGN with broadband absorption must still be embedded in the dense, ambient medium of its host galaxy. Typically, such sources are found to have jet lengths $\ll 10^1 \text{ kpc}$ ([C. P. O’Dea & D. J. Saikia 2021](#)). In Section 4.2.1, we showed that for a typical disk-like SIMBA galaxy, the

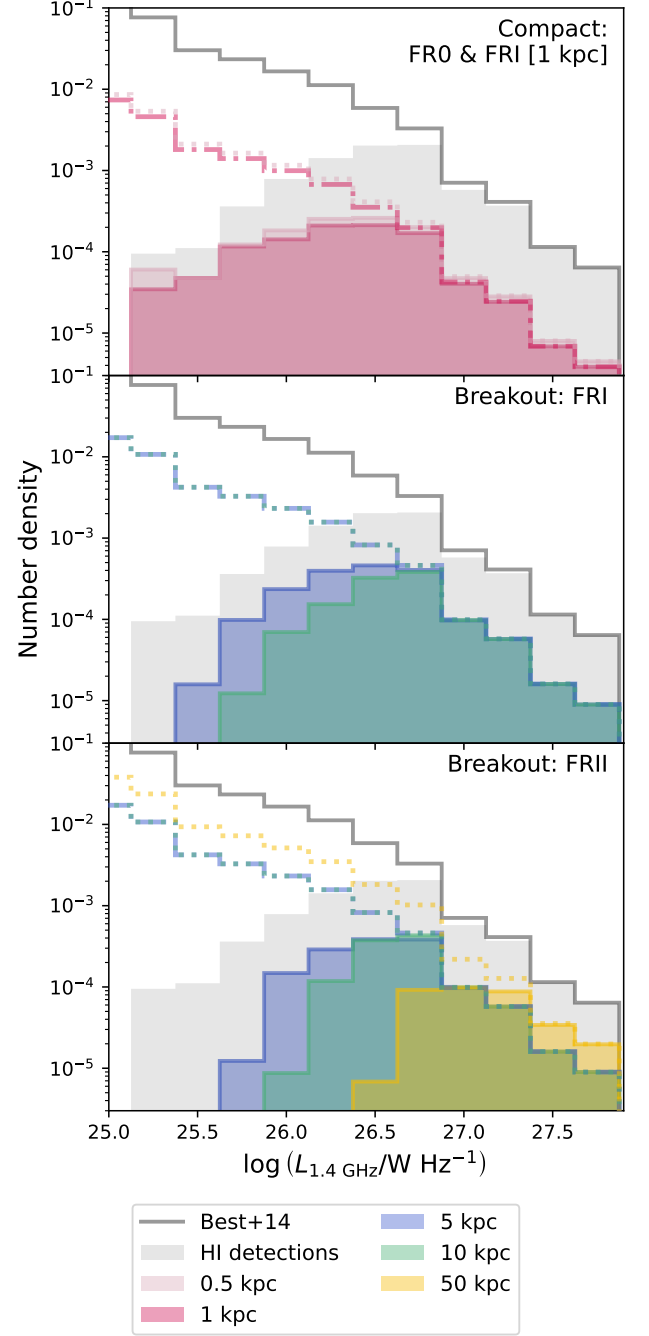


Figure 14. Population weighting of the H I detections. The solid, gray line in each subplot shows the [P. N. Best et al. \(2014\)](#) radio luminosity function for $0.5 < z < 1$ in the luminosity range of our sample. The gray, shaded region shows the luminosity distribution of all H I detections after weighting by this function. This total population is split into the different jet types, with the overall re-weighted population for each jet type given by the lines, and the H I detected population after re-weighting shown by the shaded histograms, with colours and line styles as in Figure 13. The lowest luminosities still preferentially select out compact jets, but the number density of high luminosity detections is tapered by the application of the observed luminosity function.

Jet Type	Fraction [%]
<i>Compact</i>	
FR0 [0.5 kpc]	7
FRI [1 kpc]	6
<i>Breakout</i>	
FRI [5 kpc]	14
FRII [5 kpc]	14
FRI [10 kpc]	14
FRII [10 kpc]	14
FRII [50 kpc]	31

Table 3. The ratio of different jet morphologies and lengths used as the fiducial parent population in our analysis. The compact fraction (FR0s and 1 kpc FRIs) is set to 13 per cent to match the observational constraints from the ASKAP-FLASH pilot sample (E. F. Kerrison et al. 2025). The ratio of breakout FRIIs to FRIs is set to 2:1, based on LOFAR data from M. A. Horton et al. (2025) and ASKAP data from R. P. Norris et al. (2025), where we place an additional flux density constraint $S \geq 30$ mJy on the latter to match ASKAP-FLASH constraints.

optical depth strength τ_{int} drops rapidly for jet lengths beyond ~ 1 kpc (see especially Figure 6). Therefore, we consider here any jets with lengths ≤ 1 kpc in our simulations to be still confined to their host galaxy (specifically, the FR0 and FRI-1 kpc models).

Finally, just as we specify the parent luminosity distribution of our jets, we also need to specify the parent length distribution, as our simulations currently contain an equal number of each jet length-morphology combination. Again, we focus here on the confined/breakout dichotomy, since this is most central to our comparison with ASKAP-FLASH. Observational samples put the fraction of confined jets anywhere between about 4.5 to 30 per cent of the general radio population (J. R. Callingham et al. 2017; C. P. O’Dea 1998), but we follow E. F. Kerrison et al. (2025) as our reference in this work, as their sample of compact AGN was initially selected at ~ 0.8 GHz using ASKAP survey data and a flux density limit of $S_{0.8\text{GHz}} \geq 5$ mJy. Their sample is therefore the most well-matched to the samples observed by ASKAP-FLASH. They found a compact fraction of 12.9 per cent (with a 95% confidence interval [11.7%,14.2%]). The remaining 87 per cent of the population is distributed equally amongst the remaining lengths, but with a weighting of 2:1 for FRII:FRI type jets. This ratio of FRII:FRI type jets is reflected in the observational samples from both LOFAR (M. A. Horton et al. 2025) and ASKAP surveys, provided we apply a flux density cut to their data comparable to the cut used for ASKAP-FLASH ($S \geq 30$ mJy) (R. P. Norris et al. 2025). The length distribution used for re-weighting is

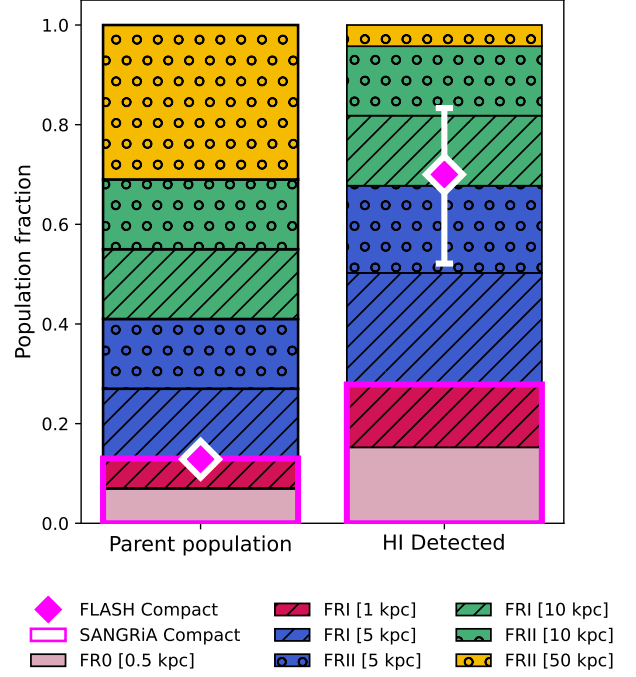


Figure 15. *Left:* The distribution of jet lengths and morphologies after re-weighting our full ‘Jet rotation’ sample by the observational constraints. The total fraction of sources classified as compact is highlighted by the thick, magenta outline. The filled diamond indicates the fraction of compact (PS) sources amongst the general radio AGN population as found in E. F. Kerrison et al. (2025). *Right:* The distribution of jet lengths and morphologies amongst our HI detections. The total fraction of compact sources is again given by the thick, magenta outline. The filled diamond indicates the fraction of HI detections in the ASKAP-FLASH Pilot Survey which are compact (PS). For both sides, the uncertainty on the observational fraction of PS sources is expressed as the 95 per cent confidence interval using a Wilson score interval. The fraction of compact sources HI detected our SIMBA sample is far lower than the compact fraction in the ASKAP-FLASH sample.

listed in Table 3. However, the exact length distribution of jets within the confined and breakout categories (not the split between confined/breakout itself) is the main source of uncertainty in our re-weighting scheme, and we consider some alternatives in Appendix A.

5.2. Geometry alone cannot explain HI detections

After fixing the parent luminosity distribution, length (compactness) distribution, and our threshold for what constitutes a breakout jet, we are finally ready to consider the HI detection fraction amongst different sub-populations of our jets. This is shown in Figure 15, where the left-hand bar gives the parent population of radio AGN after the population re-weighting discussed in Section 5. It is divided into the different jet

length/morphology combinations used in this work. The relative fractions of each of these populations have been fixed according to the ratios in Table 3. For clarity, we highlight the sources considered ‘compact’ in our simulated sample with a thick, magenta outline (both FR0s and the 1 kpc FRIs). This fraction should be directly compared to the filled diamond, which is the fraction of compact sources derived from observations with a 95 per cent confidence interval derived using a Wilson score interval. In this left-hand bar, the two are identical by construction. In the right-hand bar, we give the population fractions within the subset of radio AGN which are H I detected using the ASKAP-FLASH criterion $\ln(B) \geq 30$. Again, the compact jets in our simulation are highlighted by the thick, magenta outline, and should be directly compared to the filled diamond, which represents the fraction of compact jets detected so far in ASKAP-FLASH.

With our current assumptions, compact sources comprise only ~ 30 per cent of our mock ASKAP-FLASH detections. This is in contrast to the results from the ASKAP-FLASH survey, where 70 per cent (21/30 95% CI [52.1%,83.3%]) of sources show a clear peaked spectrum, interpreted as their being towards non-broken-out jets (E. F. Kerrison et al. 2024; H. Yoon et al. 2025). If we consider only those sources in the Pilot Survey which are classified as associated detections using the logistic regression of S. J. Curran (2021) (as was done in H. Yoon et al. 2025), the fraction of associated ASKAP-FLASH detections which are not broken out is even higher, at 12/14 (~ 86 per cent). This suggests that geometry alone is not enough to explain the high detection rate towards peaked spectrum sources. Instead, they most likely reside in galactic hosts which are uniquely rich in the cold, neutral medium traced by H I absorption. For reference, we show in Appendix A, Figure 16 the composition of the population required to match the observational fraction of H I detections towards compact sources, and it is only possible if we construct a jet population entirely from the two extremes of morphologies - compact FR0s and 50 kpc FRIIs. Also for reference we supply Figure 17, applying the same re-weighting scheme to our ‘Galaxy Rotation’ simulations, which shows a modest boost to the compact detection fraction that is still insufficient to match observations. We consider the implications of this result more fully in the following section.

6. DISCUSSION

We can now directly address the original motivation of this work: small AGN jets are only $\sim 10\%$ of the population but ~ 70 per cent of ASKAP-FLASH Pilot

Survey detections. Can this high detection rate amongst compact sources be explained by a solely geometric argument that for any jet or galaxy orientation, smaller jets are more likely to be observed in absorption because they will always be backlighting a high-column density region of the galaxy?

The observational evidence is not (yet) sufficient to answer this question. Perhaps closest is the work of Y. M. Pihlström et al. (2003), who first identified an anti-correlation between jet size and H I column densities derived from H I absorption. They argued that this relationship was due to a radially decreasing gas density profile, so that longer jets probe statistically less dense regions of the ISM. However, they did not draw any conclusions about whether this gas density profile is *different* in compact-jetted sources compared to galaxies with extended radio jets. S. J. Curran et al. (2013) built on this work, implementing simple models based on the size and geometry of both the radio continuum emission and typical H I clouds to explore the observed anti-correlation between H I column and jet size. They concluded that this relationship could be fully explained by geometry alone, even in extended sources ($l_{\text{jet}} \geq 1$ kpc). However, while geometry might explain the observed anti-correlation, their models were not tested against *detection rates* for compact and extended sources, because they were comparing with observational samples in which this was unremarkable.

Recently though, R. Morganti et al. (2026) argued that the high H I detection rate towards compact AGN jets is likely a combination of geometry and their gas-rich environment. Their analysis was focused on a targeted, observational sample of 17 radio AGN, out of which two detections of associated H I were made, both towards AGN with compact jets. Their interpretation is supported by our findings here.

To our knowledge, this work is the first test of geometrical effects on the detectability of H I absorption using cosmological simulations. In designing our experiment, we intentionally decoupled the ISM conditions and the properties of our injected radio jets, so that any difference in detection rate must be *purely* geometric. By failing to match the observed compact fraction, we have ruled out the purely geometric hypothesis, showing that jet properties and ISM conditions must be linked, which is to say that longer jets are, on average, hosted by H I-poor galaxies and/or that compact jets are found in galaxies which are preferentially H I-rich. The compact geometry of the jets alone provides only a factor of 2 increase to the detection rate against these sources, but this is still far from the factor of ~ 6 required to match the observations.

Although it is beyond the scope of this experiment to determine why compact jets could be preferentially found in H I-rich host galaxies, there are a number of interesting possibilities which other simulations are well-placed to explore. Observations show that compact jets are capable of driving multiphase outflows, including those comprising cold atomic and molecular gas, over parsec to kiloparsec scales (e.g. [R. C. Vermeulen et al. 2003](#); [M. E. Jarvis et al. 2019](#); [R. Morganti et al. 2023](#); [A. Girdhar et al. 2024](#)). This suggests radio jets could expel some fraction of nuclear H I as they propagate out to kiloparsec scales, making galaxies with extended jets potentially H I-poor. Idealised simulations have also shown that jet-ISM interactions can drive outflows, but that their properties are highly dependent on the nuclear gas density, the jet power, the direction of the jets relative to the galactic gas disk, and local gas turbulence ([D. Mukherjee et al. 2018](#); [R. Tanner & K. A. Weaver 2022](#); [O. Borodina et al. 2025](#), and see review by [D. Mukherjee 2025](#)). Therefore both observations and simulations suggest that breakout jets may have evacuated hosts of gas (ejective feedback). In some cases the resulting turbulence may also fact drive inflows back onto the AGN ([D. Mukherjee et al. 2018](#)).

Alternatively, if the galaxies which host compact radio jets are H I - or more generally gas-rich, it is reasonable to think that the broadband spectral turnover (which is the observational signature of their compactness) might also be caused by free-free absorption in this dense, circumnuclear medium (e.g. [G. V. Bicknell et al. 1997](#)). Under this assumption, the idealised simulations of [S. A. Young et al. \(2025\)](#) can not only explain the broadband spectral signature of compact radio jets, but also provide a sensible model for the evolution of their spectrum as they propagate out into the less-dense, outer regions of their host galaxy.

These considerations of jet-ISM interactions remain beyond the scope of this work because of limitations in our current set up. First of all, the spatial resolution of SIMBA is not high enough to adequately resolve AGN-driven outflows at the level where their multiphase composition could be mapped, as is possible in the idealised simulations mentioned above. Furthermore, the atomic hydrogen content of the gas in SIMBA is determined via sub-grid models, and although these allow for a two-phase ISM that is calculated on-the-fly, the ‘cold’ gas is typically at $\sim 10^3 < T < 10^5$, still at least an order of magnitude hotter than the atomic hydrogen traced observationally with H I absorption ([R. Davé et al. 2020](#)).

As we discussed in Section 2.3, our approach is to impose a spin temperature of $T_s = 300$ K when calculating the 21 cm cross section, and allow the line to be broad-

ened by the kinetic temperature of the gas particle. This acts as an effective thermal and kinetic broadening component to account for unresolved turbulence below the resolution of SIMBA and generates absorption lines of depths and widths that match well to those observed with ASKAP-FLASH. To further investigate these assumptions, we could run the SANGRIA pipeline on simulations with an explicitly resolved cold ISM, such as the idealised simulations mentioned above, or the FIRE suite ([P. F. Hopkins et al. 2018](#)). This would allow a more detailed study of the H I kinematics and absorption line profiles, albeit for a much smaller sample of galaxies.

Finally, we do not link our jet parameters to the BH properties recorded in the SIMBA catalogues by choice, but by doing so we lose any information about possible jet-driven outflows, implemented in SIMBA as kinetic-mode AGN feedback. Likewise, the luminosity distribution of our jets, while physically motivated, is simplistic. Coupling jet length to luminosity could be achieved by making use of pre-established semi-analytic models that track these properties over the lifecycle of AGN jets (e.g. [M. J. Hardcastle 2018](#); [R. J. Turner et al. 2023](#)). And ideally, this could all be joined with insights from high-resolution, idealised simulations like those mentioned above could be used to inform the next iteration of 21 cm absorption mocks from cosmological simulation.

There are several interesting options for future work building on our setup though which would allow us to address at least some of the limitations in the current work, and which could be pursued independently or combined in a number of useful ways. Perhaps the most natural is to link our injected jet properties with the BH properties of the SIMBA galaxies, following the prescriptions of [N. Thomas et al. \(2021\)](#) where appropriate. This would also be an opportune way to explore whether compact jets are preferentially launched in gas-rich environments within cosmological simulations, linking back to our findings here. For an even more careful match to ASKAP-FLASH, a larger study of H I absorption in the full SIMBA box, incorporating both associated and intervening detections, would be most useful. Doing so would, however, limit the number of bright radio jets in the box, as discussed in Section 3.3. Nevertheless, such an approach would no doubt also benefit from more physically motivated jet modelling also, incorporating semi-analytic approaches like those of [M. J. Hardcastle \(2018\)](#) and [R. J. Turner et al. \(2023\)](#). These modifications would also pave the way for a full mock lightcone, that stack together portions of the SIMBA box from different redshifts to emulate a full, 3 dimensional observation, as [N. Thomas et al. \(2025\)](#) produced for a mock

observation of the MIGHTEE survey (M. Jarvis et al. 2018). Such mocks then open up many possibilities for studying cosmological evolution in simulated galaxies.

7. CONCLUSIONS

In this work, we have applied SANGRIA, a new framework for mock 21 cm absorption spectra, to a collection of AGN-hosting galaxies in the SIMBA cosmological simulations. Motivated by observational findings, especially those of the ASKAP-FLASH Pilot Surveys, our aim was to test whether H I absorption is preferentially detected towards compact radio jets due to geometry alone. For this reason, we intentionally decoupled our injected radio jets from the BH properties of our chosen SIMBA galaxies, with the result that any observed variation in detection rates could be purely attributed to jet and galaxy geometry, rather than jet-ISM interactions.

We demonstrate that our pipeline reproduces expected results from ASKAP-FLASH when varying the physical parameters of the jet-galaxy configuration. The strength of H I absorption scales inversely with jet length for a given luminosity because the same amount of radio flux density is spread over a wider area (Section 4.2.1). The significance of H I features also scales with the luminosity, as this is effectively increasing the signal to noise on detections (Section 4.1). Overall the strongest lines appear when a galaxy is viewed edge-on (Section 4.3), but jet orientation has a minimal affect when averaged across the full sample, except for the longest, 50 kpc jets (Section 4.4). Furthermore, as seen in the ASKAP-FLASH Pilot Surveys (Sadler et al. *in prep*), H I line profiles are wider in bulge-dominated galaxies than their disk-dominated counterparts (Section 4.5). We also find that our line profiles are widest in systems undergoing active merging, strong interactions or which have an extended, asymmetric gas distribution, suggesting that these galaxies have the most kinematically disturbed gas. We suggest that such a population of complex galaxies traced by wide absorption profiles will likely be detected in larger samples of H I lines from the full ASKAP-FLASH survey.

Putting all these findings together, in Section 5 we modify the distribution of our simulated jet-galaxy configurations to better match observed samples. We weight the whole population such that the luminosity distribution follows the observed luminosity function of radio galaxies at $z \sim 0.5$. We additionally weight different jet configurations so that our distribution of jet morphologies matches both the fraction of compact/broken-out morphologies in the literature, and the observed ratio of FRI:FRII type jets. As a result, we can directly compare the fraction of compact jets in our semi-realistic

population of mock H I detections to the same fraction in the ASKAP-FLASH Pilot Survey. We find that even with the conservative assumption that all detections in the ASKAP-FLASH Pilot Survey are of H I gas physically associated with the radio source, geometric configurations alone cannot reproduce the observed detection rate towards compact sources within our framework. We interpret this result as evidence that compact, peaked spectrum sources must be significantly more gas rich than the hosts of extended radio jets in order to produce such a high detection rate in an untargeted survey like ASKAP-FLASH.

In this paper we presented a framework to compare new large observational surveys to cosmological simulations in an apples-to-apples manner which can give us insight into the physical connection between AGN jets and their galaxy hosts - namely that pre-breakout AGN are more likely to reside in gas-rich galaxies. With larger observational datasets, SANGRIA can go far beyond the use in this paper and directly test the cold gas fraction in cosmological simulations beyond the local Universe, probe observational biases in calculating the observed the cosmic H I mass density, Ω_{HI} , and make predictions about the high- z H I forest (an analogue to the Ly α forest) that may soon be detectable with the SKA.

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APPENDIX

A. ALTERNATIVE POPULATION COMPOSITIONS

In Section 5.2, we show the relative fractions of the jet population in our entire, re-weighted ‘Jet rotation’ sample, as well as the relative fractions of AGN which are H I-detected. To test the effect of our parent population weights more rigorously we consider a few alternative models here.

Figure 16 provides three alternative parent populations and the resulting H I detected population, again drawn from the ‘Jet rotation’ simulations. On the left, we assume a simple, two-component population comprising 13 per cent FR0s, and 87 per cent 10 kpc FRIIs. This represents a scenario where the compact, FR0 jets remain confined for some non-negligible amount of time, and then evolve rapidly upon initial breakout from the host galaxy before reaching a slowly evolving state at a length of approximately 10 kpc. As a result, the vast majority of the observed population are either in their compact phase or a fully extended phase. This model also assumes that more extended structures, like the 50 kpc jets, are at a low surface brightness and are not detected in the ASKAP-FLASH observations. The resulting detection rate towards compact sources (25%) is not appreciably different to the rate we find in Section 5.2 where the extended jets are more evenly distributed amongst different lengths. The same assumptions hold for the middle panel of Figure 16, except we allow for some of the 50 kpc jets to be luminous enough for detection in the ASKAP-FLASH observations. This boosts the compact source fraction of H I detections to 36 per cent, but it is still significantly outside the 95 per cent confidence interval for the observed compact H I detection rate.

The third and final alternative considered here is that the rapid expansion phase takes place over even larger length scales, so that the observational sample can be described by only compact, FR0s, and 50 kpc FRIIs, the most extended jets in our simulations. This approximately mimics the size distribution of M. A. Horton et al. (2025). Under this assumption, it is possible to recover the high rate of H I detections towards compact jets (70 per cent) from geometric arguments alone. However, observational samples show that this is not a good characterisation of the length distribution of radio galaxies; the M. A. Horton et al. (2025) is biased towards large galaxies because they require an angular size (> 90 arcsec) for galaxies in their sample to be resolved by LOFAR. By comparison, M. A. DiPompeo et al. (2013) examine a number of samples and show that they contain exponentially more compact sources than extended (> 50 kpc) ones. Further, in the R. P. Norris et al. (2025) ASKAP sample, 88 per cent of sources have angular sizes < 90 arcsec, though at present their catalogue does not have accompanying redshift information with which to derive linear sizes, this is planned for a future work.

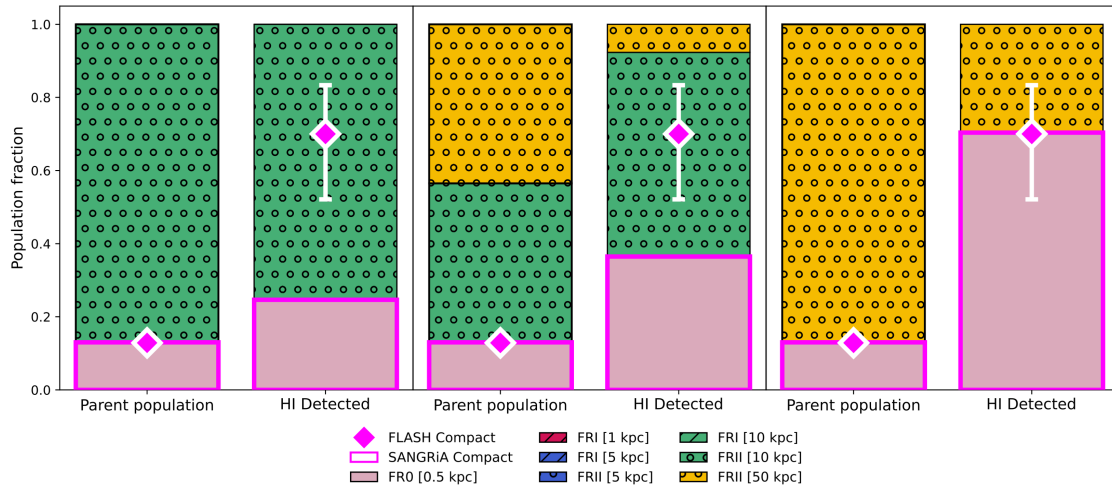


Figure 16. Predicted detection fractions for simplified jet populations, assuming only FR0 (compact) and FRII (breakout) jets. The compact fraction is set to 13% (E. F. Kerrison et al. 2025). The assumed length of the breakout jets are varied across the three panels: *left*: all 10 kpc; *middle* 50:50 10 kpc and 50 kpc; *right*: all 50 kpc. Only in the extreme case of assuming all breakout jets are 50 kpc in radius does the predicted compact HI detection fraction match the observed value. This demonstrates that our conclusion that such high compact fractions cannot be achieved by inclination alone holds in all but the most extreme assumptions about the parent jet population.

B. ROTATING JET AND GALAXY

For completeness, we include here a version of Figures 5 and 13 for our ‘Galaxy rotation’ suite of simulations, where the jet and galaxy are fixed perpendicular to one another, then rotated through 180 degrees. These are shown in Figure 17. Overall, the same trends seen in the ‘Jet rotation’ simulations discussed in the main text are seen here again; compact sources contribute the vast majority of H I detections at low ($L_{1.4\text{GHz}} < 10^{26} \text{ W Hz}^{-1}$) radio luminosities, but the overall fraction of compact sources amongst H I detections does not reach that seen in ASKAP-FLASH Pilot Survey observations.

The biggest difference is for the 50 kpc FR II jets – there are now some geometries where the H I is never detected, no matter how bright the luminosity. This is reflected in the dotted yellow line in the left-hand panel never reaching 1. This has a small effect on the detected fractions (right panel), pushing up the detected compact fraction in the rightmost bar chart. However, this effect only increases the compact fraction from 27 per cent to 34 per cent which is still significantly less than the observed value of 73 per cent. Thus our conclusion that geometry alone cannot explain this observed compact fraction still holds even in this alternative rotation scenario.

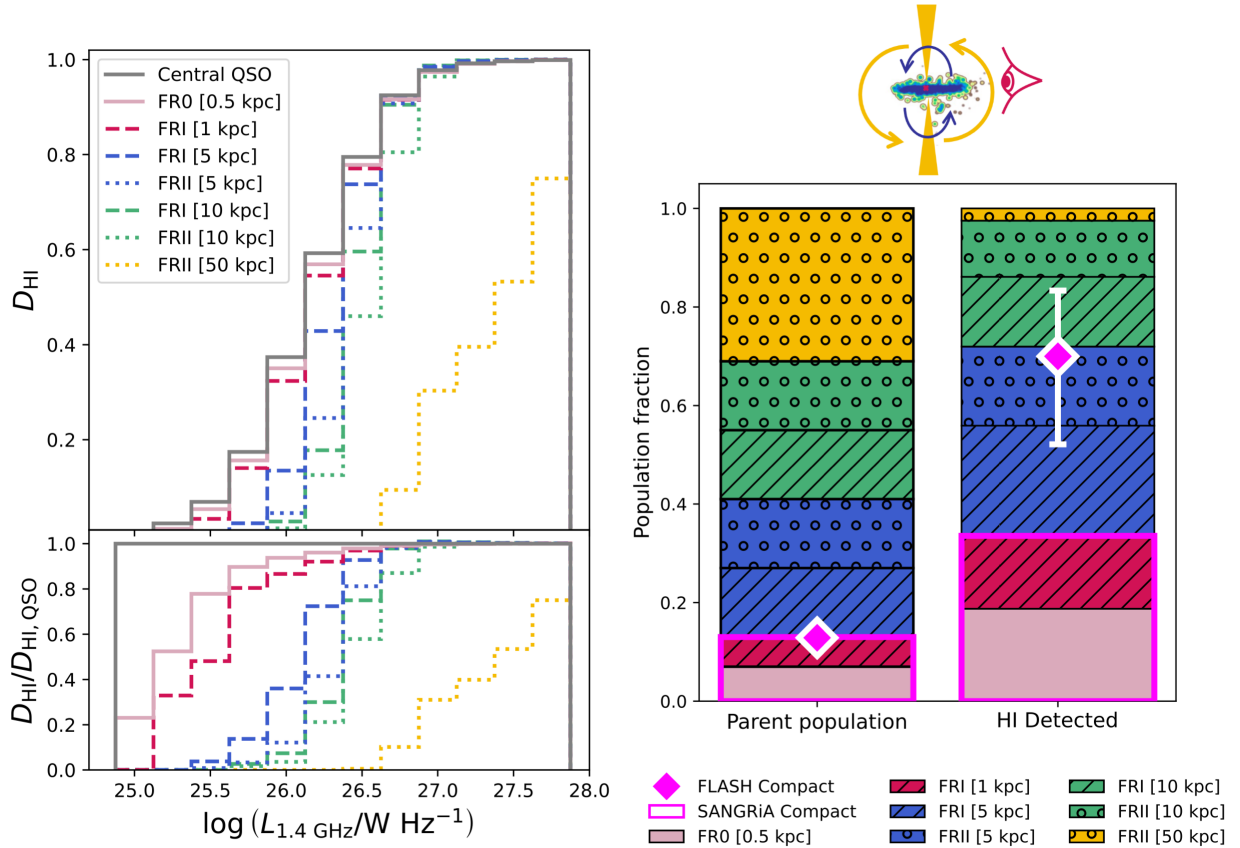


Figure 17. *Left:* The H I detection fraction as a function of luminosity (see Figure 13) and *Right:* The ratio of jet types detected in H I compared to the overall parent population (see Figure 15). These are now calculated for ‘Galaxy rotation’ simulations, and the results are broadly similar to ‘Jet rotation’ case presented in the main body of the paper. The most significant differences are seen in the 50 kpc FR II detection rate at high luminosities, and the overall detection rate of compact sources.